Camunda 8 - Getting Started with Human Workflow

Business Requirement

We will be using a fictitious organization; Camundanzia, as the scenario for this course.

Camundanzia

Scenario

The Camundanzia insurance company would like to introduce process automation throughout their organisation. The first process they would like to automate is their **Request Annual Leave** process.

Employee annual leave requests must be submitted to the employee's **Line Manager**. The **Line Manager** must review the request and then either approve it or reject it with a justification. Currently, the request is made via email.

User Story

Two simple User Stories to capture Camundanzia's requirements can be found below:

User Story

- As an **Employee**
- I want to submit my annual leave request for approval by my **Line Manager**
- So that I get an official response.

User Story

- As a **Line Manager**
- I want to approve or reject annual leave requests made by my **Employees**
- So that I can effectively manage the workload of my team

Acceptance Criteria

We will take an MVP approach to developing an application that meets these requirements.

The initial acceptance criteria that we will be working to are:

1. **Employees** are able to submit an annual leave request
2. **Line Managers** can review annual leave requests
3. **Line Managers** can approve, reject or ask for more details on a request
4. **Employees** can provide additional information on a request and then resubmit
5. **Line Managers** can review additional information on a request before making a decision

Iterative Development

Iterative Development

Camunda is well suited to an iterative development approach; applications can be easily revised and upgraded without affecting in-flight processes.

Outline Solution

We now have a clear set of requirements and acceptance criteria for the Camundanzia *Request Annual Leave* process.

The sections below will provide you with an overview of how to develop a **Human Workflow** solution that meets these requirements.

Camundanzia

Scenario

Imagine that you have recently joined Camundanzia as a new Developer and have been assigned to the Camunda project team. Your first assignment is to implement the *Request Annual Leave* process using the requirements from the previous lesson.

Fortunately for you, a more experienced Camunda Developer has drafted an outline solution for you. Your task is to implement a solution based upon the outline that has been provided.

Outline Solution

For each of the acceptance criteria that have been identified, the more experienced Developer has provided a summary of the Camunda components that provide the implementation of the requirements in addition to the configuration that is required by you, the Developer.

Submit Annual Leave Request

The core of the solution is provided by being able to execute a **Process Instance** in Camunda. In order to execute a **Process Instance** we need to create a **Process Definition** which models the business process.

- **Process Definitions** are created using the Modeler
- **Process Instances** are executed using the Zeebe Engine
- **Process Instances** can be monitored using Operate

Create Process Definition

Create a **Process Definition** that represents the *Request Annual Leave* process

Create Request Annual Leave Task

Create a **Task** which can be used by the **Employee** to submit their request.

Create Request Form

Create a **Form** which can be used by the **Employee** to enter the details of their request.

Review Annual Leave Request

Users interact with Camunda using the Tasklist. Tasklist provides OOTB functionality for viewing tasks that have been assigned to a User, claiming tasks so that they can be worked on and completing tasks so that the execution of the process can continue.

Create Review Request Task

Create a **Task** which can be used by the **Line Manager** to review a request.

Approve or Reject the Request

Users can add or update the data in a process through the use of **Forms** which are exposed through **Tasklist**.

Create Review Form

Create a **Form** which can be used by the **Line Manager** to make a decision on the request.

Provide Additional Information & Resubmit

Process Definitions can contain BPMN constructs such as **Gateways** which control the flow of the business process. **Gateways** use **Process Variables** which have been populated in advance to define the next step to take in the process.

Create Provide Additional Information Task

Create a **Task** which can be used by the **Employee** to re-submit their request with a comment.

Create Gateway

Create an **Exclusive Gateway** which can be used to conditionally control the process flow.

Development Plan

Now that we have identified the components and features of Camunda which will be used to implement our solution, we can create a plan of the work that needs to be performed and the tools that will be required.

- **Create a Process Definition**
  - This will be configured using the Modeler
- **Create Forms**
  - These will be configured using the Modeler
- **Deploy the Process Definition**
  - This will be performed from the Modeler
- **Launch a Process Instance**
  - This will be performed from the Modeler
- **Test the Process Definition**
  - This will be performed from Tasklist and Operate

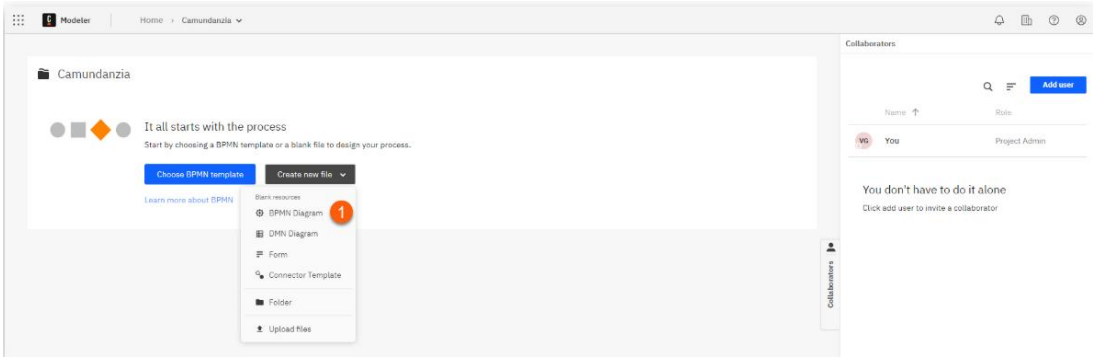
Create Process

We will now create a new **Process Definition** (BPMN Diagram) in our **Project** which will be used to model the *Request Annual Leave* process.

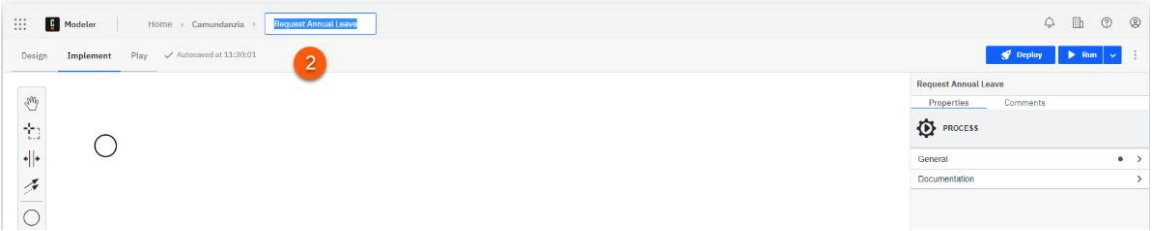
Create Process Definition

Create a new **Process Definition** by following the steps below:

1. In the **Modeler**, click the **New** button and then **BPMN Diagram** from the menu that appears

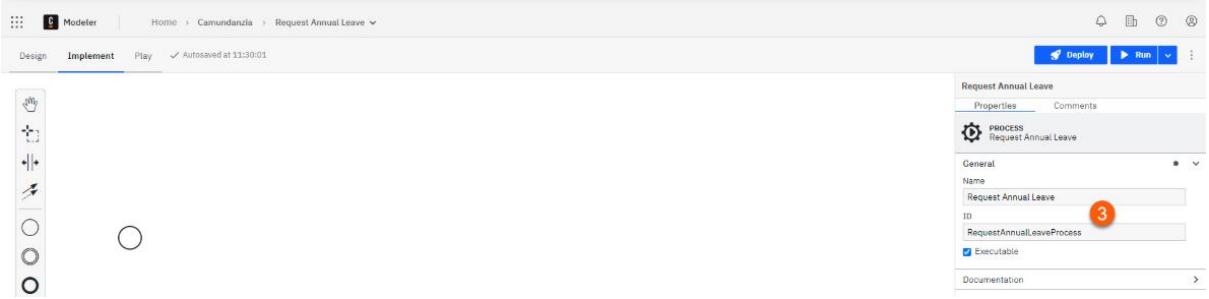


2. This will open the Modeler with a new **BPMN Diagram**. Click on the **Diagram Name** and then **Edit Name** from the menu that appears to change the name of the BPMN Diagram. For this course we will use *Request Annual Leave*.



3. **Make sure you are in the Implement tab on the top left corner of the screen.** In the properties panel on the right-hand side of the screen, set the **Name** and **ID** as follows:

Name	Value
Name	Request Annual Leave
ID	RequestAnnualLeaveProcess



Naming Conventions for BPMN IDs

Define developer-friendly and business-relevant IDs for the process itself, as well as all activities, messages, and errors. Also consider events, gateways, and the sequence flows that carry conditional expressions. Even though IDs are just identifiers, keep in mind that they will show up regularly on the technical level. Meaningful IDs will help a lot.

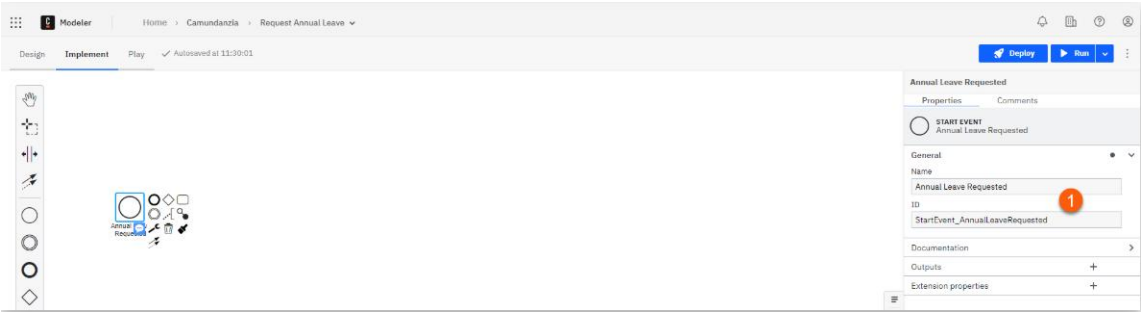
Model Business Process

Add Start and End Events

All **Process Definitions** require at least one **Start Event** and at least one **End Event**.

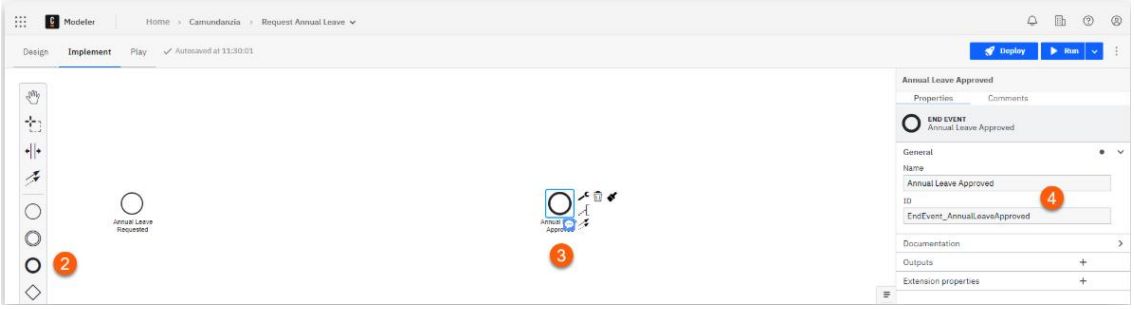
1. Click on the **Start Event** in the process and then expand the **General** tab in the properties panel on the right-hand side of the screen. Set the **Name** and **ID** as follows:

Name	Value
Name	Annual Leave Requested
ID	StartEvent_AnnualLeaveRequested



2. Click on the **End Event** from the **BPMN Palette** to add a new element to the diagram.
3. Click on the right-hand side of the diagram to place the **End Event**.
4. Set the **Name** and **ID** for the **End Event** as follows:

Name	Value
Name	Annual Leave Approved
ID	EndEvent_AnnualLeaveApproved



Naming Events

Wherever possible, name an event using an object and a verb reflecting a state. Always try to describe which state an object is in when the process is about to leave the event.

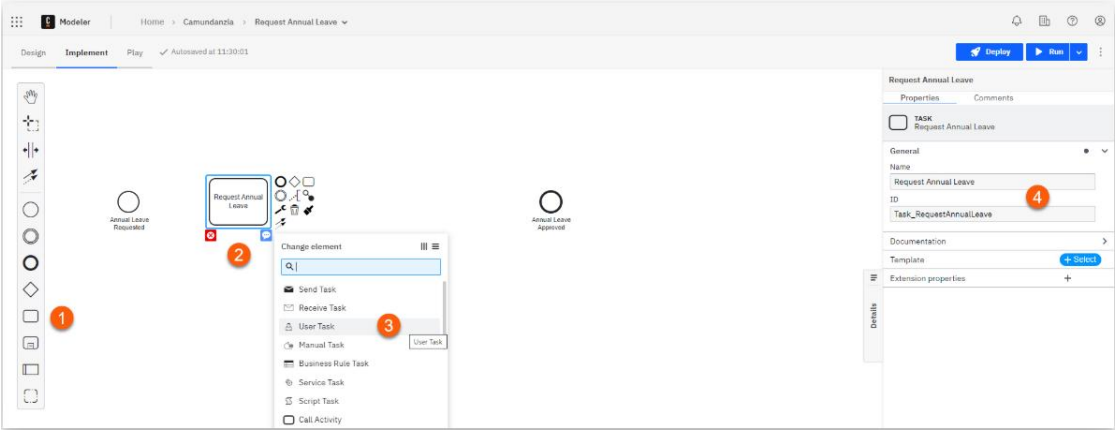
Add User Tasks

For the first draft of our process, we will create two **User Tasks**; *Request Annual Leave* and *Review Request*.

1. Click on the **Task** from the **BPMN Palette** to add a new element to the diagram.
2. Click to the right of the **Start Event** to place the **Task**.
3. Click on the **Change Type (Wrench)** icon in the toolbox and then select **User Task** from the context menu that appears

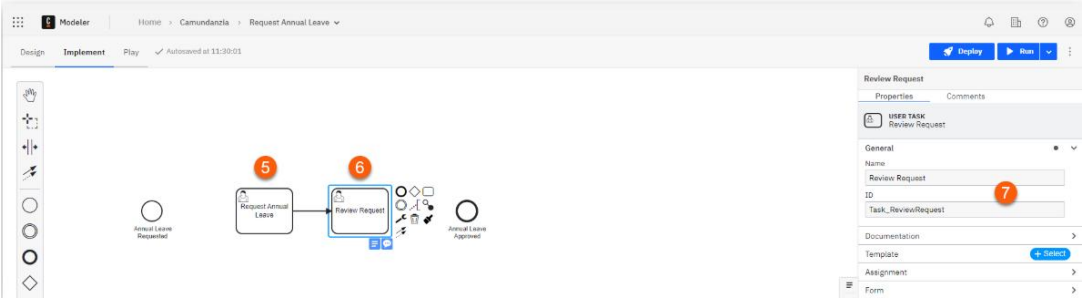
4. Set the **Name** and **ID** for the **User Task** as follows:

Name	Value
Name	Request Annual Leave
ID	Task_RequestAnnualLeave



5. Click on the *Request Annual Leave* **User Task** and then click on the **Append Task** icon in the toolbox that appears. This will automatically append a **Task** to the *Request Annual Leave* **Task** in the process.
6. Click on the **Change Type** icon in the toolbox and then select **User Task** from the context menu that appears
7. Set the **Name** and **ID** for the **User Task** as follows:

Name	Value
Name	Review Request
ID	Task_ReviewRequest



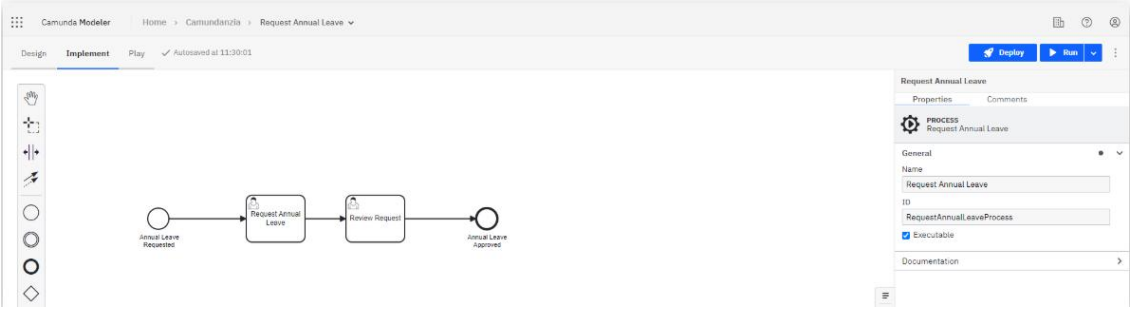
Add Task or Append Task?

Both methods used above to add **User Tasks** to the diagram result in the same outcome. Feel free to experiment with both approaches to see which you prefer.

Add Sequence Flow

We now need to connect the elements that have been added to the process so that they can be executed.

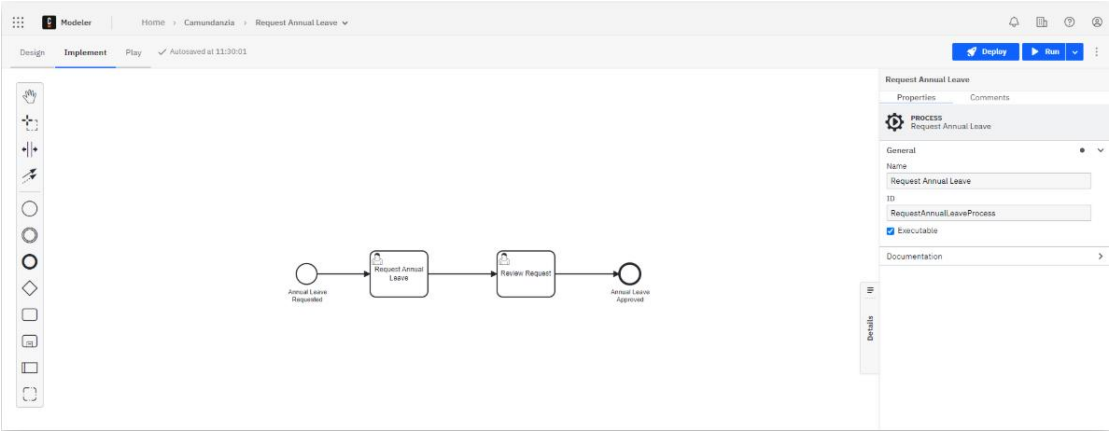
1. Click on the *Annual Leave Request* **Start Event** and then click on the **Connect using Sequence** icon in the toolbox that appears.
2. Click on the *Request Annual Leave* **Task** to create a **Sequence Flow** between the **Start Event** and **User Task**.
3. Perform the same steps to connect the *Review Request* **User Task** to the *Annual Leave Approved* **End Event**.



Arrange Diagram

Finally, we can arrange the diagram using the **Create / Remove Space Tool**.

1. Click on the **Create / Remove Space Tool** in the **BPMN Palette** on the left-hand side of the screen.
2. Use the tool to evenly space the BPMN elements on the diagram. Your diagram should look similar to the one be low when you have finished.



Readable Diagrams

We create visual process models to better understand, discuss, and remember processes. Hence, it is crucial that models are easy to read and understand.

3. The **Modeler** automatically saves your diagram after each change so there is no need to explicitly save the diagram that you have created.

Created a Process Definition

Congratulations! You have now created a **Process Definition** which captures the *Request Annual Leave* process.

Deploy Process

We will now need to deploy our **Process Definition** to the previously created **Cluster** so that we can start a **Process Instance**.

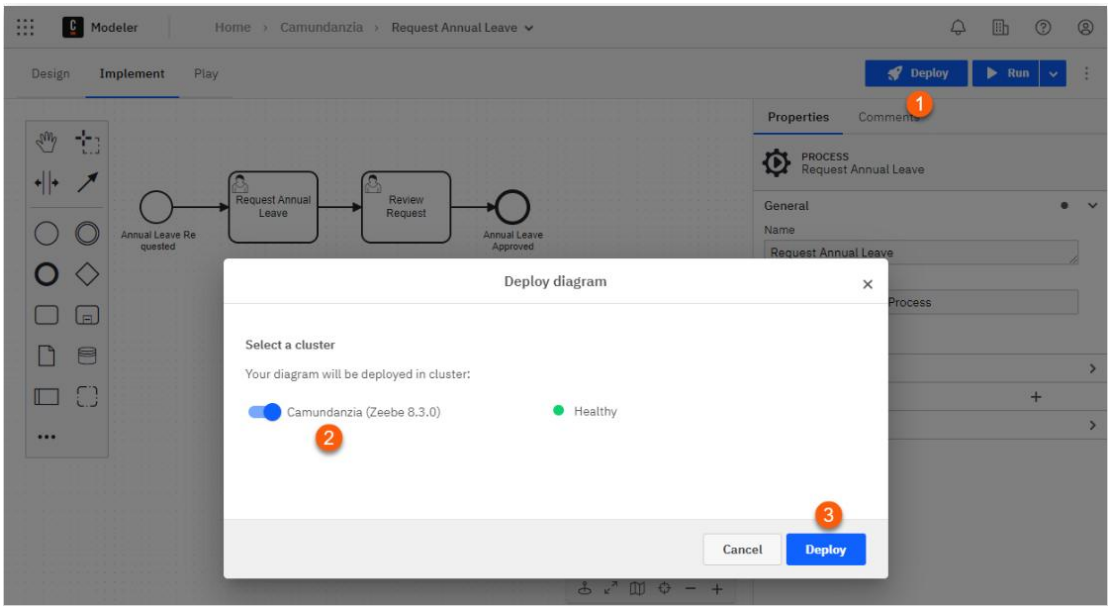
Deploy Process Definition

Deploy the **Process Definition** by following the steps below:

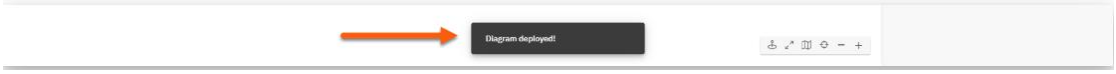
What is a Deployment?

A **Deployment** allows the **Process Definition** to be executed by the **Zeebe Engine**.

1. Click on the **Deploy** button in the top right corner of the **Modeler**
2. Ensure that the *Camundanzia* **Cluster** is selected from the dialog box that appears.
3. Click the **Deploy** button to deploy the **Process Definition** to the **Cluster**



4. A **Diagram Deployed** message will be displayed at the bottom of the **Modeler** to show that this was successful.



View the Deployment in Operate

Next, we can take a look into **Operate** to confirm that the deployment has been successful.

What is Operate?

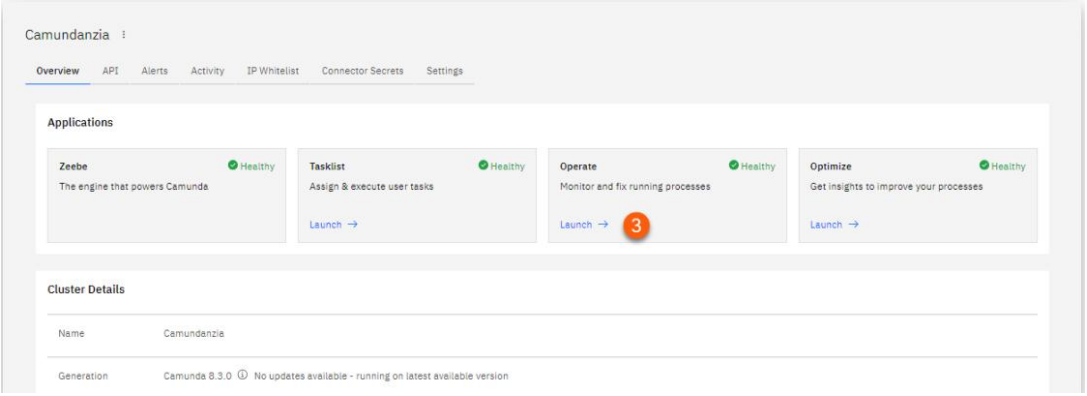
Operate is a tool for monitoring and troubleshooting process instances running in the **Zeebe Engine**.

In addition to providing visibility into active and completed process instances, Operate also makes it possible to carry out key operations such as resolving incidents, and updating process instance variables.

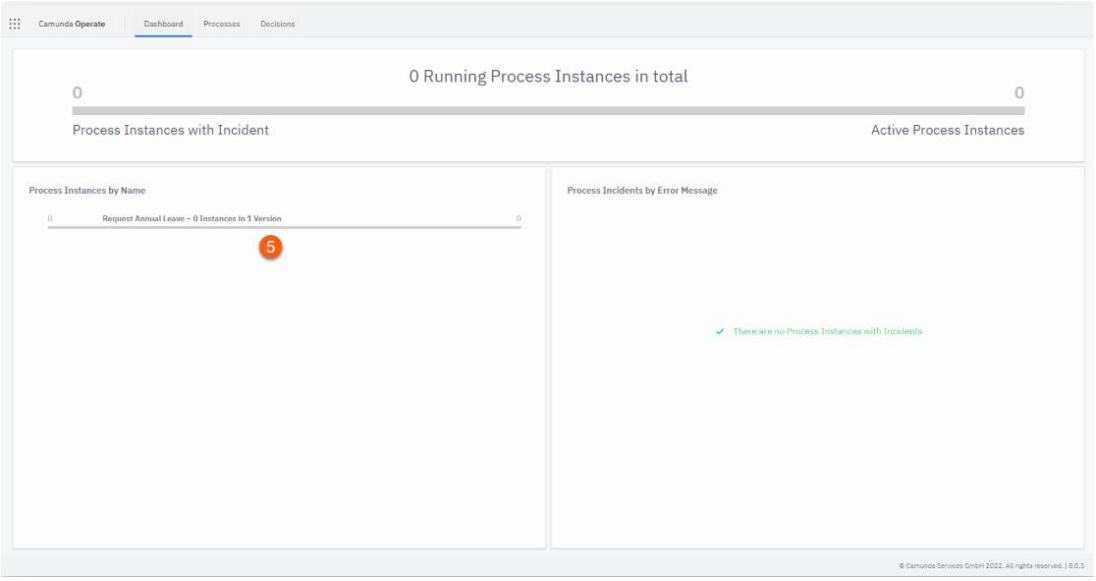
Launch Operate

Launch **Operate** by following the steps below:

1. From the **Console**, select **Clusters**.
2. Select *Camundanzia* from the **Clusters** page.
3. Click the **Launch** link next to **Operate** in the **Cluster Details** page



4. After a few seconds, the **Operate** Dashboard will be displayed.
5. You will also be able to see that the *Request Annual Leave* process is visible in the Dashboard.



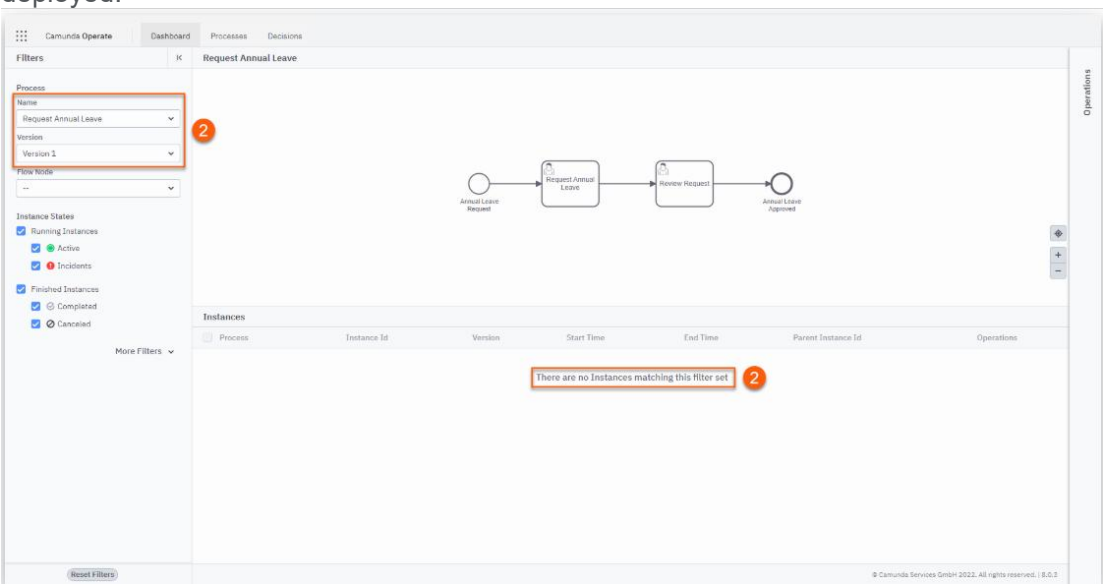
View Process Instances

We can now view the process instances in **Operate** by following the steps below:

What is a Process Instance?

One execution of a **Process Definition** is called a **Process Instance**. Many **Process Instances** can be launched from one **Process Definition**.

1. Click on *Request Annual Leave* from the **Instances by Process** panel on the left-hand side of the screen
2. **Operate** will not show any active process instances at this point. However, we can see that version 1 of our *Request Annual Leave* process has been successfully deployed.



Deployed a Process Definition

Congratulations! You have now deployed the *Request Annual Leave* **Process Definition** into the **Zeebe Engine**.

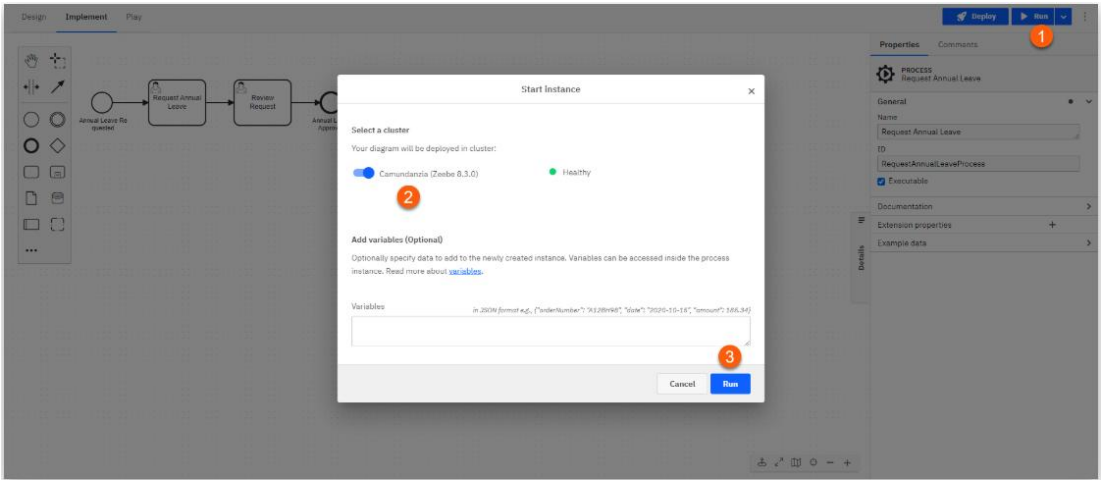
Start Process

We will now need to execute our **Process Definition** for the first time; this is also known as starting a **Process Instance**.

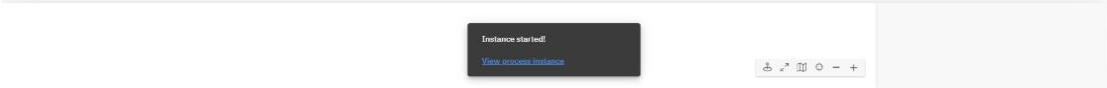
Start a Process Instance

Start an instance of the *Request Annual Leave* by following the steps below:

1. Click on the **Run** button in the top right corner of the **Modeler**
2. Ensure that the *Camundanzia* **Cluster** is selected from the dialog box that appears.
3. Click the **Run** button to start a new instance of *Request Annual Leave* using the *Camundanzia* **Cluster**



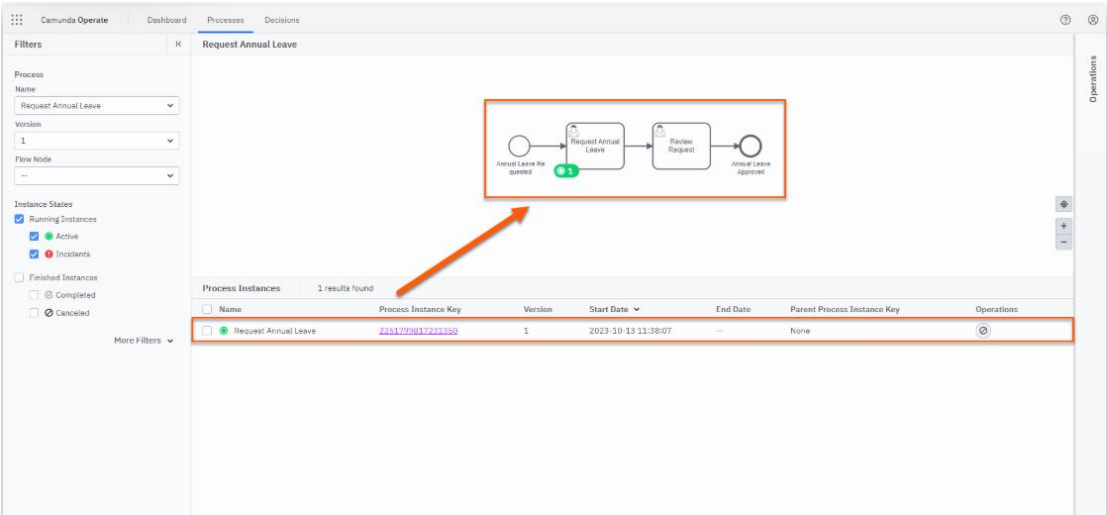
4. An **Instance Started!** message will be displayed at the bottom of the **Modeler** to show that this was successful.



View Process Instances

We can now view the process instances in **Operate** by following the steps below:

1. Launch **Operate** by following the steps in the previous lesson. Alternatively, refresh the window if you still have **Operate** open.
2. Click on *Request Annual Leave* from the **Instances by Process** panel on the left-hand side of the screen
3. **Operate** will now show a running instance of our *Request Annual Leave* process. The completion of the *Request Annual Leave* **User Task** is still pending; you can see this from the presence of the green marker against the **User Task**.



Started a Process Instance

Congratulations! You have now started an instance of the *Request Annual Leave* process.

Test Process

Finally, in order to test our *Request Annual Leave* process, we will need to complete the **User Tasks** using **Tasklist**.

What is Tasklist?

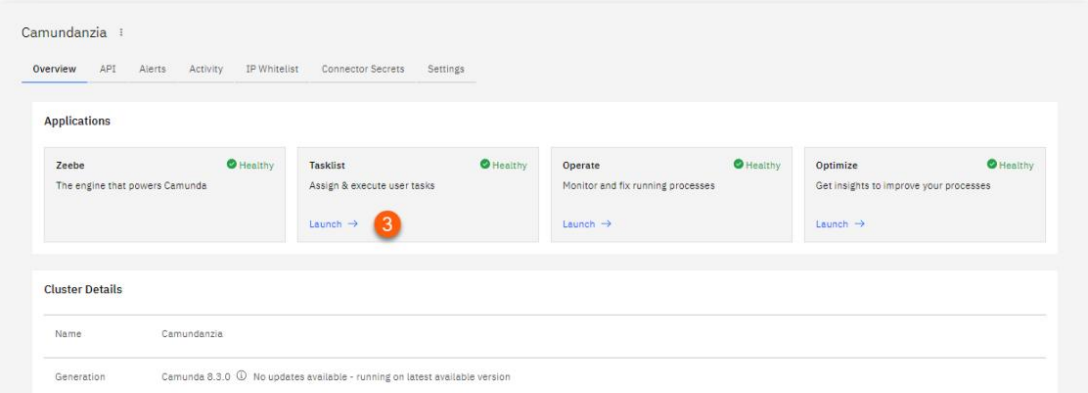
Tasklist is an out-of-the-box web application that's tightly integrated with Camunda's process orchestration capabilities. Simply model a business process and deploy it to the **Zeebe Engine**; when a user needs to work on a task, they'll see it appear in **Tasklist**.

Test the Process

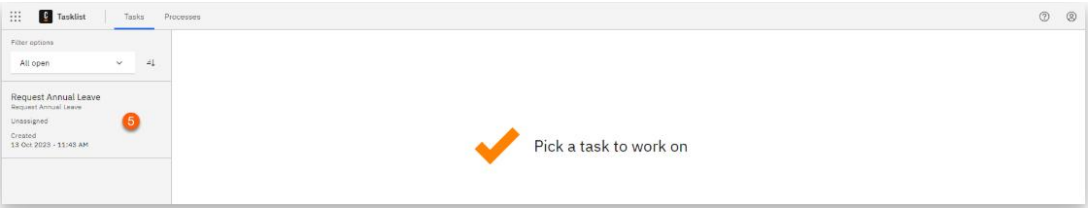
Launch Tasklist

Launch **Tasklist** by following the steps below:

1. From the **Console**, select **Clusters**.
2. Select *Camundanzia* from the **Clusters** page.
3. Click the **Launch** link next to **Tasklist** in the **Cluster Details** page



4. After a few seconds, the **Tasklist** Dashboard will be displayed.
5. You will also be able to see that a *Request Annual Leave* **User Task** is visible in the left-hand panel of **Tasklist**.



Assign the Task to you

Assign the *Request Annual Leave* **User Task** to you by following the steps below:

1. From the left-hand panel, click *Request Annual Leave*. At the moment, there is no way to **Complete** the task or provide input at this point; the task must be assigned to you first.

   Why do I need to assign a Task to me?

   Assigning a task sets the `assignee` of the task to the current user. Assigning a task allows the user to update or complete the task.

2. **Assign** the task to you to start working on it

3. Once the task has been assigned to you, you will be able to see it as **"Assigned to me"**. You also have the option to **Unassign** the task to undo the **assignment**. In addition, two more buttons have appeared, **Add Variable +** and **Complete Task**.

Complete the Task

In order for the **Line Manager** to see the dates of the request, we will need to add two variables to the process using **Add Variables +**.

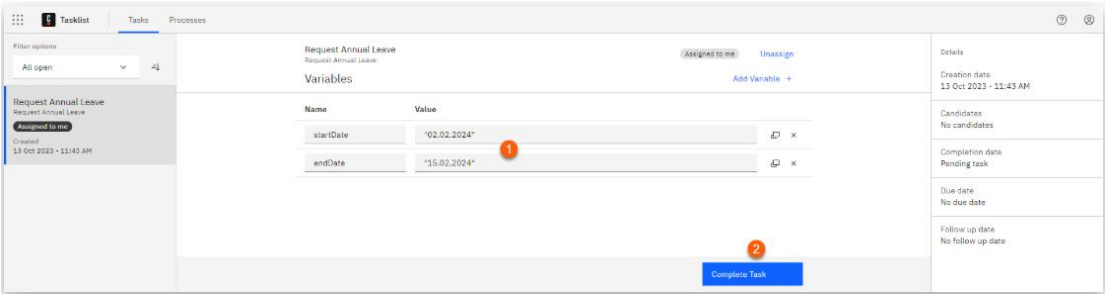
What is a Variable?

Variables are part of a process instance and represent the data of the instance.

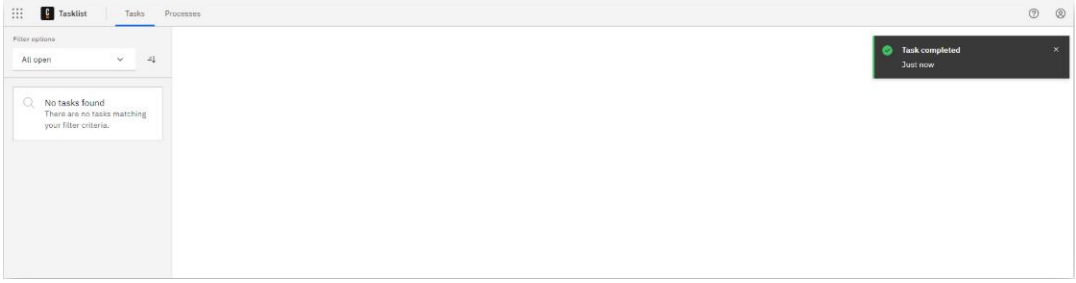
1. Use **Add Variables** to create the two variables below (don't forget the double quotes):

Name	Value
startDate	"02.02.2024"
endDate	"15.02.2024"

2. Click the **Complete Task** button to complete the task.



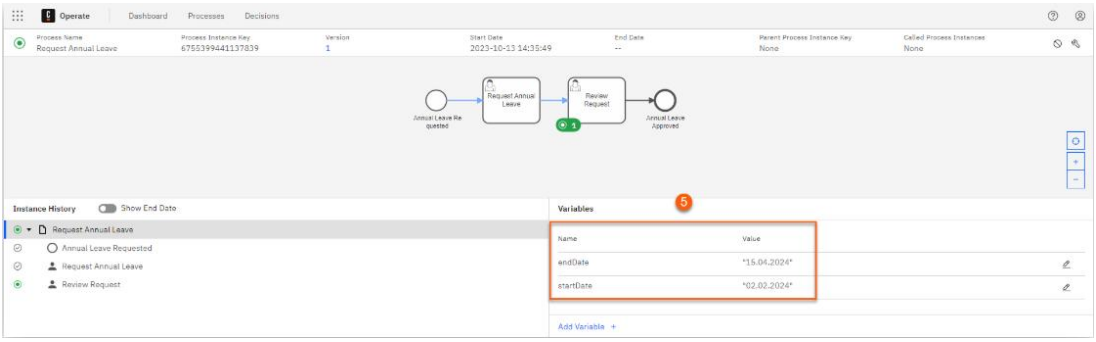
3. A **Task Complete** message will be displayed to show that this was successful.



View Completed Request Task in Operate

We can now view the completed task in **Operate** by following the steps below:

1. Launch **Operate** by following the steps in the previous lesson. Alternatively, refresh the window if you still have **Operate** open.
2. Click on *Request Annual Leave* from the **Instances by Process** panel on the left-hand side of the screen
3. **Operate** will now show a running instance of our *Request Annual Leave* process. The completion of the *Request Annual Leave* **User Task** is still pending; you can see this from the presence of the green marker against the **User Task**.
4. Click on the **Instance Id** link in the **Instances** panel to view the details of the process instance.
5. Note that the variables have now been populated for the process instance and that a green marker has appeared next to the *Review Request* task.



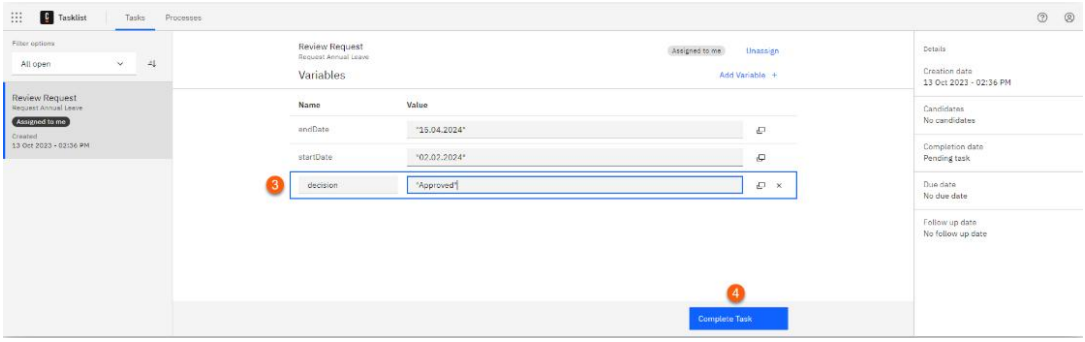
Complete the Review Task

Finally, the **Line Manager** now needs to approve the request.

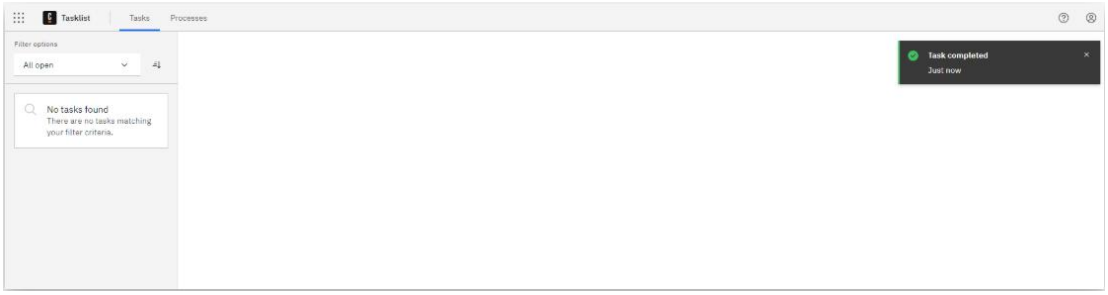
1. Refresh **Tasklist** to see a new *Review Request* task.
2. **Assign** the task to you. You will notice that the `startDate` and `endDate` variables have already been populated for the task.
3. Use **Add Variables** to create an additional variable (don't forget the double quotes):

Name	Value
decision	"Approved"

4. Click the **Complete Task** button to complete the task.



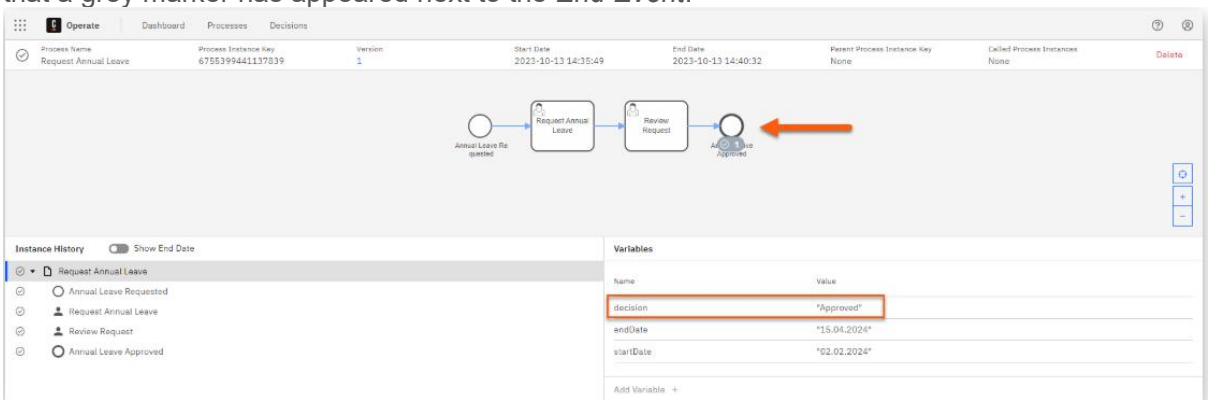
5. A **Task complete** message will be displayed at the bottom of **Tasklist** to show that this was successful.



View Completed Review Task in Operate

We can now view the completed task in **Operate** by following the steps below:

1. Launch **Operate** by following the steps in the previous lesson. Alternatively, refresh the window if you still have **Operate** open.
2. Click on *Request Annual Leave* from the **Instances by Process** panel on the left-hand side of the screen
3. **Operate** will now show a completed instance of our *Request Annual Leave* process.
4. Click on the **Instance Id** link in the **Instances** panel to view the details of the process instance.
5. Note that the `decision` variable has now been populated for the process instance and that a grey marker has appeared next to the *End Event*.



Completed a Process Instance

Congratulations! You have now completed the first instance of the first version of the *Request Annual Leave* process.

Create Request Form

Overview

We can now create User Tasks, add Process Variables and complete User Tasks. However, the only information that a User Task currently contains is the name of the task. Additionally, the variables must be created manually and can therefore be modified in subsequent tasks.

This provides a poor experience for the user as data may be omitted, data may be entered in the wrong format or data may be invalid.

What is a Form?

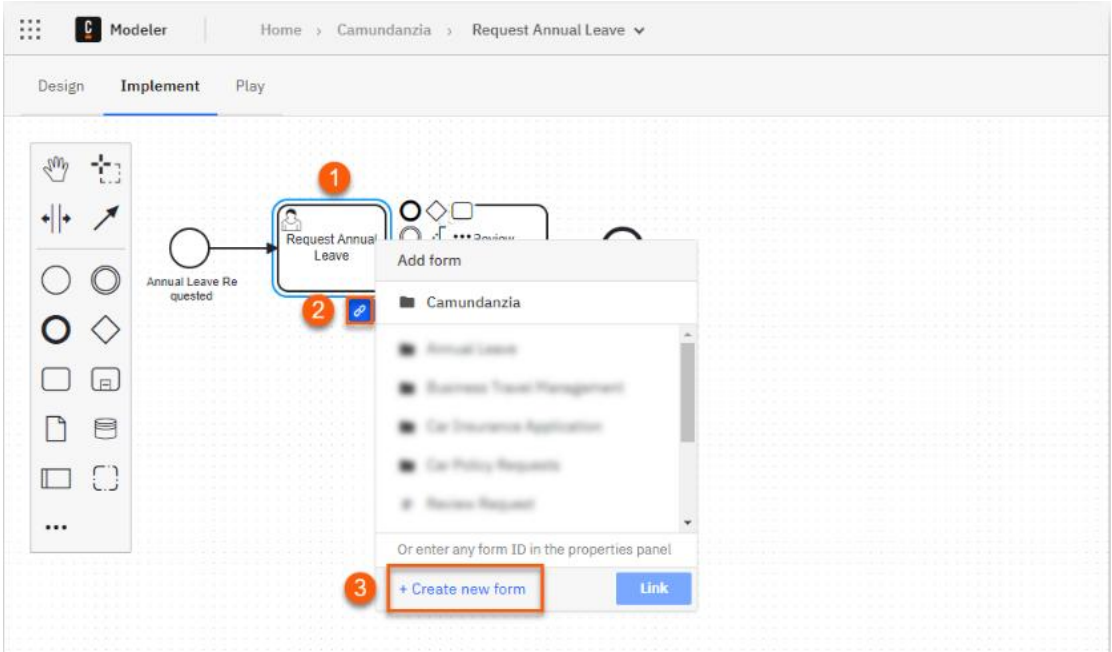
Forms are used to capture data into specially designed field types such as text, date, select and multiple choice radio buttons. They are used where human intervention is required within a process and this intervention is handled by filling in a form that is displayed in a user's task list using the **Tasklist** application.

We will now create a new **Form** in our **Project** which will be used to interact with the **Request Annual Leave** task.

Create Form

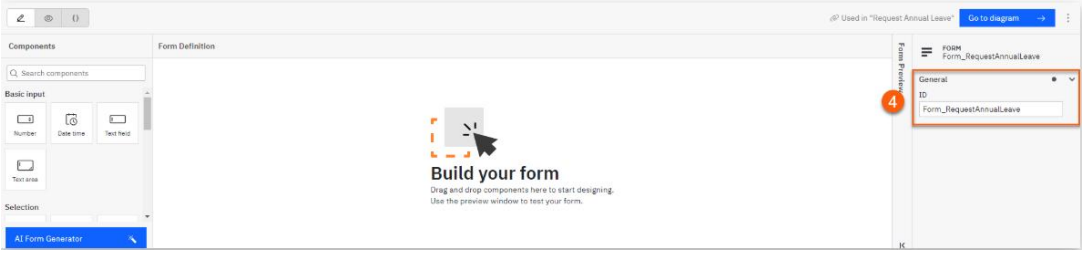
Create a new **Form** by following the steps below:

1. Select the **Request Annual Leave** task
2. Click the icon to add a form
3. Click **Create new form**



4. In the properties panel on the right-hand side of the screen, set the **ID** as follows:

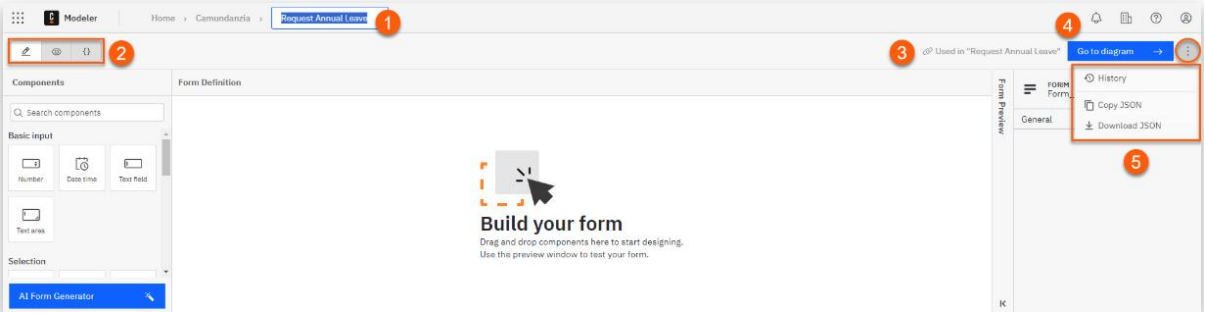
Name	Value
ID	Form_RequestAnnualLeave



Now your form is created in your Project.

Keep this in consideration:

1. It took the name of the human task automatically
2. Different modes are available: *design*, *validate* and code *editor*
3. It displays where this form is used
4. You can directly go to the diagram that is using this form
5. More options are available from the *three-dots* icon



Design Form

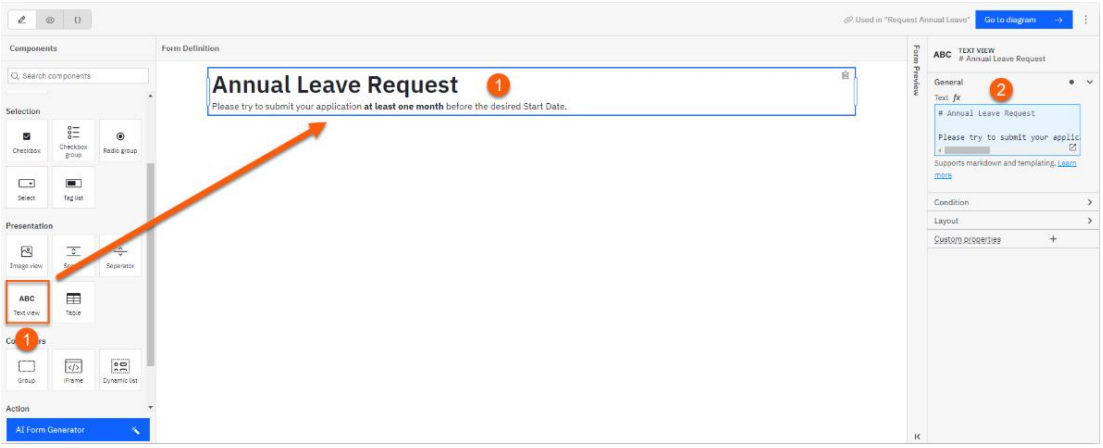
On the left-hand side of the **Form Designer**, you will find the different **Form Elements** that can be used in a **Form**.

By using Drag and Drop, you can select an element that should be displayed in your **Form**.

Design your **Form** by following the steps below:

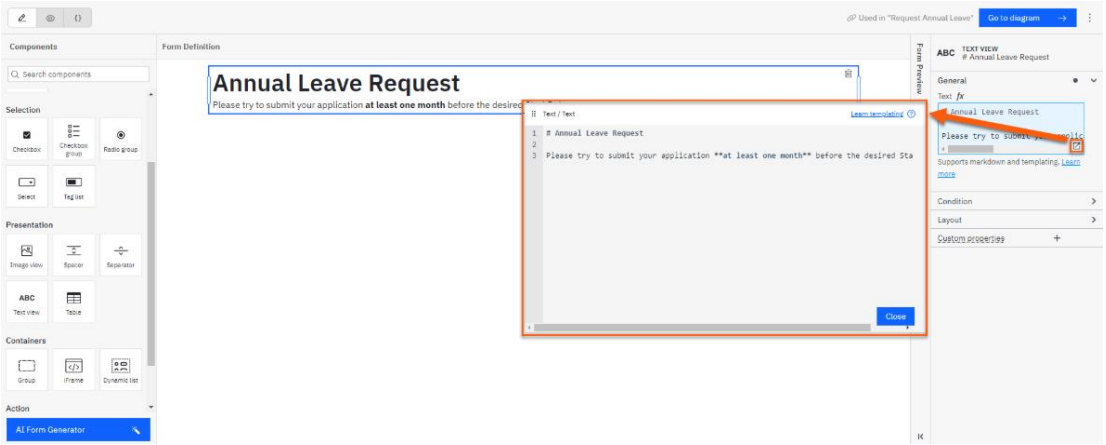
1. Select the **Text view** element and drag it into the centre of the **Form**. This will be used to provide instructions to the **Employee**
2. Select the **Text view** element on the **Form** and configure it by using the **Properties Panel** on the right-hand side of the screen.

Name	Value
Text	# Annual Leave Request Please try to submit your application **at least one month** before the desired Start Date.



Open pop-up editor

You can open the pop-up editor by clicking the icon that appears on *mouse over* the text field area



Markdown or HTML

The **Text view** element allows for text to be displayed using either Markdown or HTML.

- Markdown Cheatsheet
- HTML Cheatsheet

3. Select the **Date time** and drag it into the centre of the **Form**, below the **Text field** element. This will be used to enter the **Start Date** of the request.
4. Select the **Date time** on the **Form** and configure it by using the **Properties Panel** on the right-hand side of the screen.

Section	Name	Value
General	Date Label	Start Date
General	Key	startDate
Validation	Required	True

Additional Properties

The **Date time** contains considerably more properties than the **Text view** element as this is used to capture input from a user in addition to displaying text.

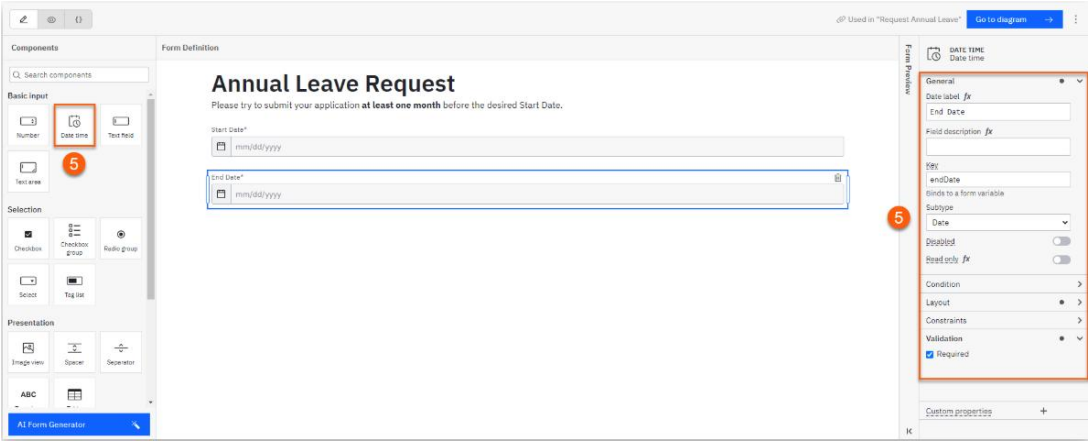
Key

It is important to use a meaningful name for the **Key** field. The **Key** field maps to a **Process Variable** in the **BPMN Diagram** and is likely to be referenced in subsequent forms or BPMN constructs.

5. Add another **Date time** with the same configuration as above and the following value for **Date Label** and **Key**.

Section	Name	Value
General	Date Label	End Date
General	Key	endDate
Validation	Required	True

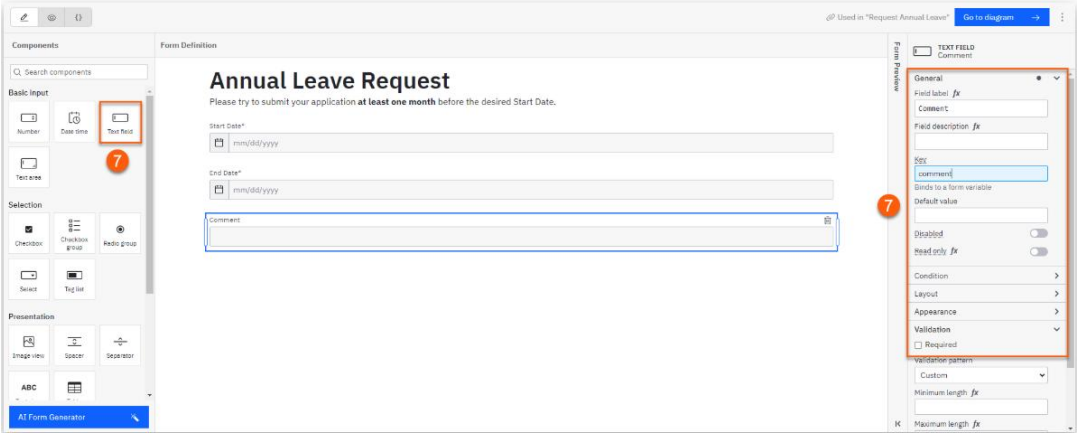
6. Your **Form** should now look similar to the below:



7. Finally, add an additional **Text Field** with the following configuration:

Section	Name	Value
General	Field Label	Comment
General	Key	comment
Validation	Required	False

8. Your completed **Form** should now look as follows:



Created a Form

Congratulations! You have now created a **Form** which can be used to interact with the *Request Annual Leave* User Task.

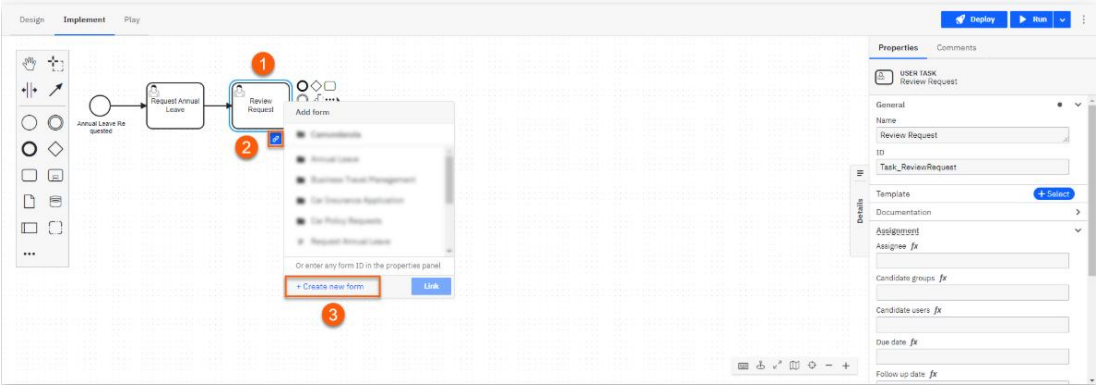
Create Review Form

Next, we will now create a new **Form** in our **Project** which will be used to interact with the **Review Request** task.

Create Form

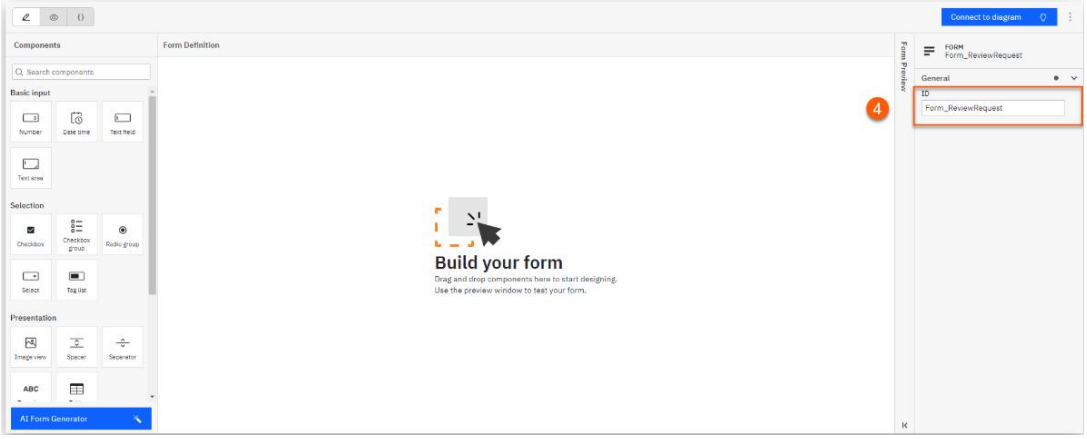
Create a new **Form** by following the steps below:

1. Select the **Review Request** task
2. Click the icon to add a form
3. Click **Create new form**



4. In the properties panel on the right-hand side of the form editor, set the **ID** as follows:

Name	Value
ID	Form_ReviewRequest



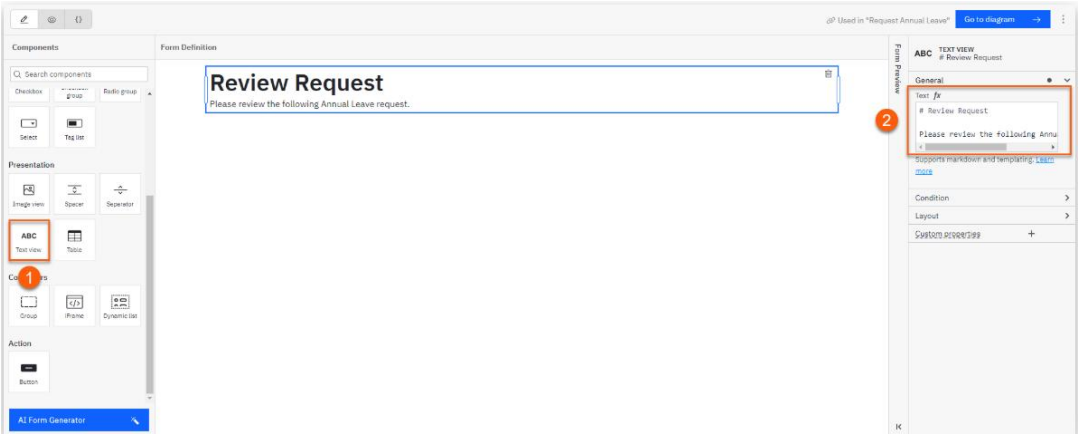
Now your form is created in your Project.

Design Form

Design your **Form** by following the steps below:

1. Select the **Text view** element and drag it into the center of the **Form**. This will be used to provide instructions to the **Line Manager**
2. Select the **Text view** element on the **Form** and configure it by using the **Properties Panel** on the right-hand side of the screen.

Name	Value
Text	# Review Request Please review the following Annual Leave request.



3. Select the **Date time** and drag it into the centre of the **Form** below the **Text view** element. This will be used to enter the **Start Date** of the request.
4. Select the **Date time** on the **Form** and configure it by using the **Properties Panel** on the right-hand side of the screen.

Section	Name	Value
General	Date Label	Start Date
General	Key	startDate
General	Read only	True

Read only

The **Review Request** form needs to be read-only to prevent accidental editing. For this reason, ensure that the **Read only** property is set to `True`.

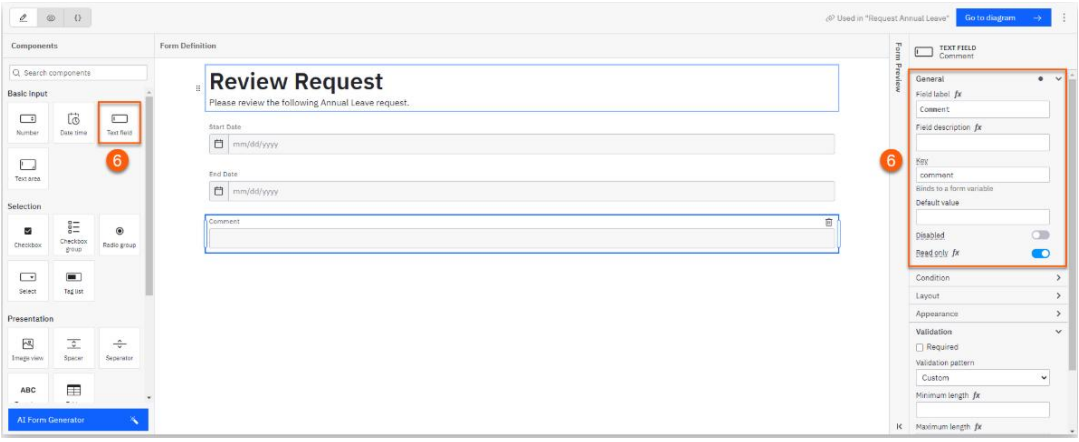
5. Add another **Date time** with the following values.

Section	Name	Value
General	Date Label	End Date
General	Key	endDate
General	Read only	True

6. Add a **Text Field** with the following values.

Section	Name	Value
General	Field Label	Comment
General	Key	comment
General	Read only	True

7. Your **Form** should now look as follows:



8. Finally, add a **Radio** element with the following configuration:

Section	Name	Value
General	Field Label	Decision
General	Key	decision
General	Read only	False
Validation	Required	True

Required

If the *Decision* is not set as `Required`, the user may forget to select one of the options. This will result in problems later in the process as it will not be clear which activity should be performed next. To avoid this scenario, ensure that the checkbox for *Required* is selected, in the Validation section.

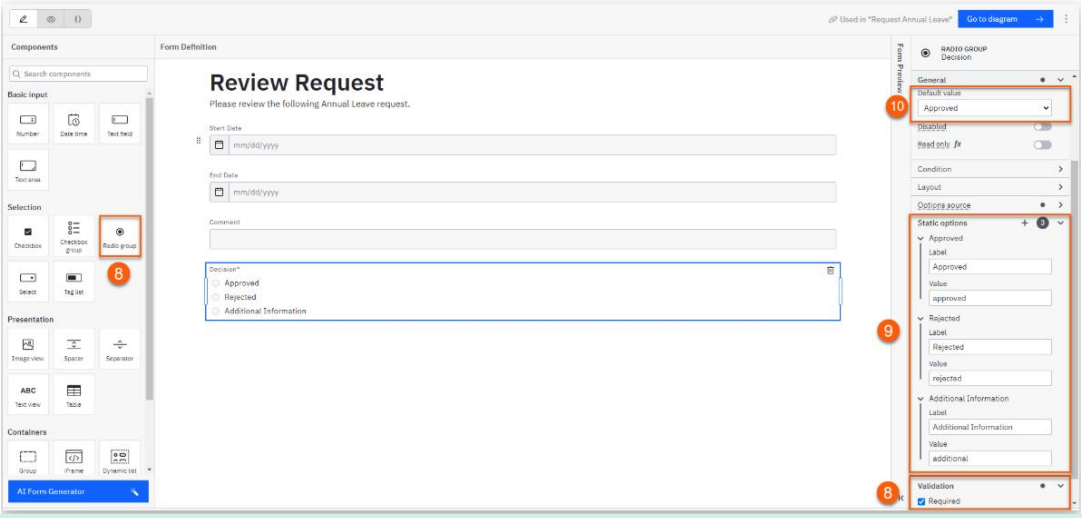
9. **Radio** elements also require a series of key-value pairs to represent the options. Configure the options as follows:

Section	Label	Value
Static options	Approved	approved
Static options	Rejected	rejected
Static options	Additional Information	additional

10. Now select the *Default value*:

Section	Name	Value
General	Default Value	Approved

11. Your completed **Form** should now look as follows:



Created a Form

Congratulations! You have now created a **Form** which can be used to interact with the *Review Request* User Task.

Deploy Process

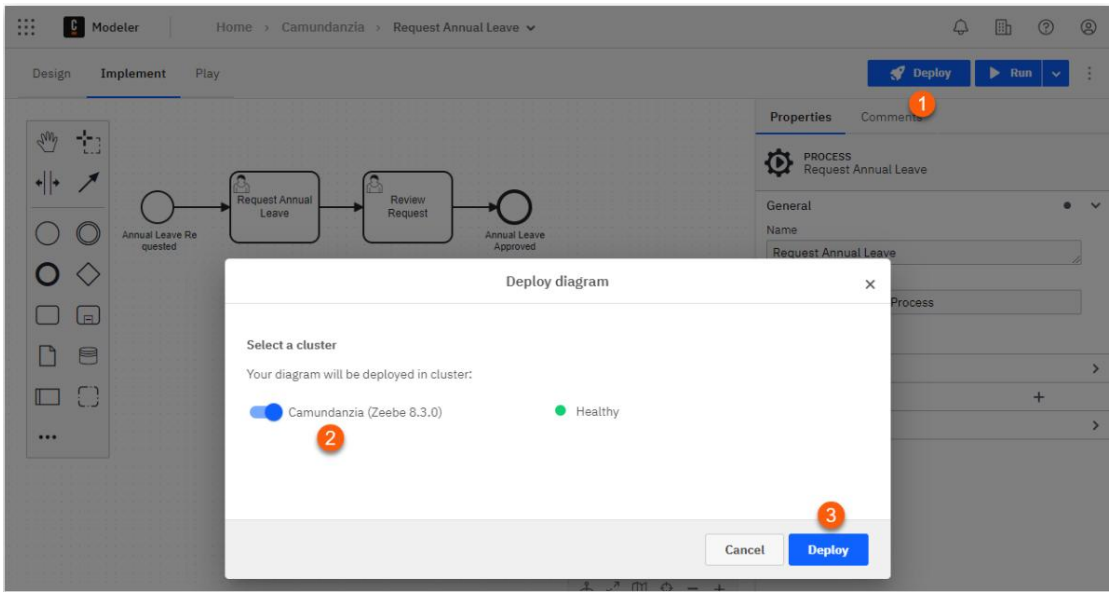
We will now need to deploy our updated **Process Definition** to the previously created **Cluster** so that we can start a new **Process Instance** against the updated version of the **Process Definition**.

Deploy Process Definition

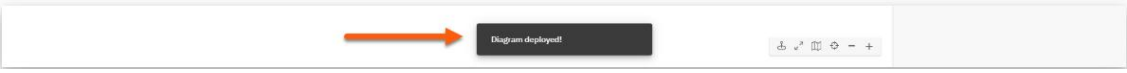
Make sure you are in the Implement tab on the top left corner of the screen.

Deploy the **Process Definition** by following the steps below:

1. Click on the **Deploy** button in the top right corner of the **Modeler**
2. Ensure that the *Camundanzia* **Cluster** is selected from the dialog box that appears.
3. Click the **Deploy** button to deploy the **Process Definition** to the **Cluster**



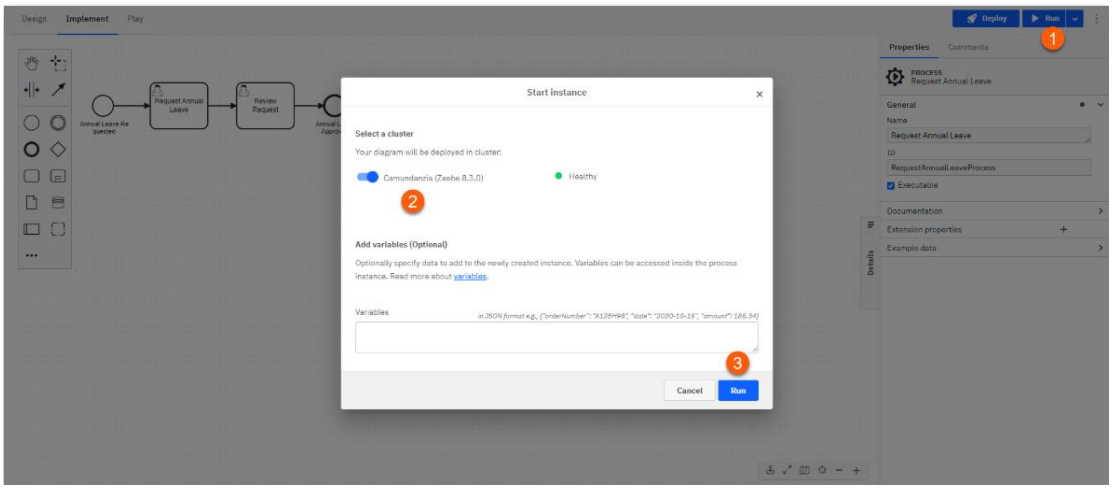
4. A **Diagram Deployed** message will be displayed at the bottom of the **Modeler** to show that this was successful.



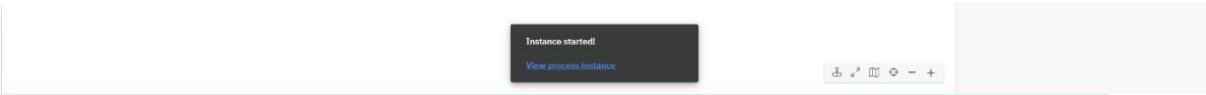
Start a Process Instance

Start a new instance of the updated *Request Annual Leave* process by following the steps below:

1. Click on the **Run** button in the top right corner of the **Modeler**
2. Ensure that the *Camundanzia* **Cluster** is selected from the dialog box that appears.
3. Click the **Run** button to start a new instance of *Request Annual Leave* using the *Camundanzia* **Cluster**



4. An **Instance Started!** message will be displayed at the bottom of the **Modeler** to show that this was successful.



Deployed an updated Process Definition

Congratulations! You have now deployed and started the updated *Request Annual Leave* **Process Definition** which now contains **Forms**.

Test Process

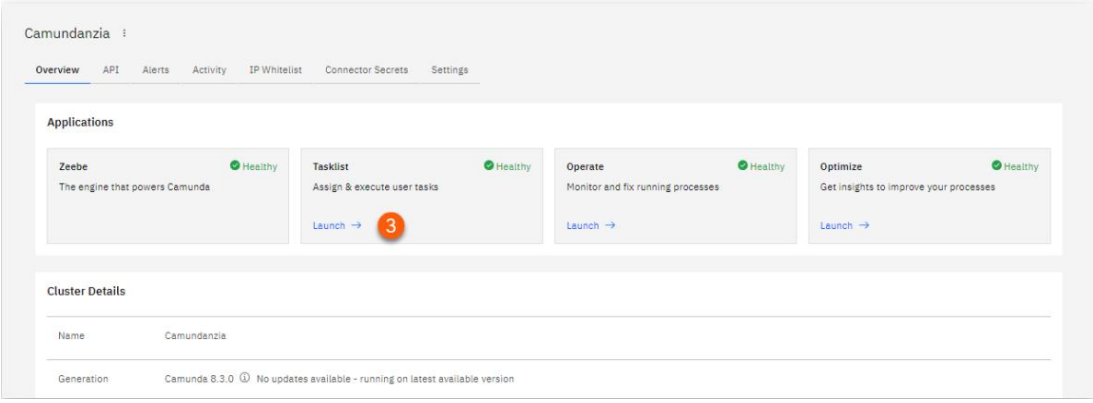
Finally, in order to test our updated *Request Annual Leave* process, we will need to complete the User Tasks using **Tasklist**.

Test the Process

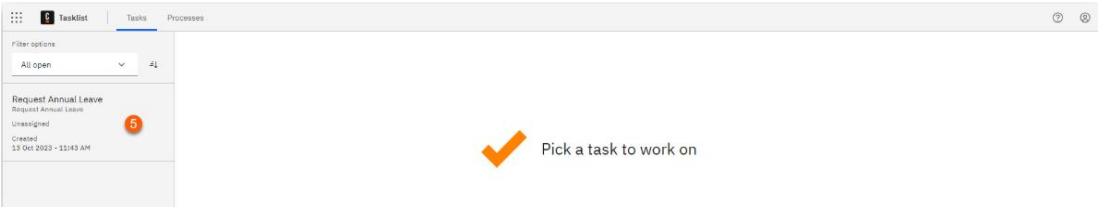
Launch Tasklist

Launch **Tasklist** by following the steps below:

1. From the **Console**, select **Clusters**.
2. Select *Camundanzia* from the **Clusters** page.
3. Click the **Launch** link next to **Tasklist** in the **Cluster Details** page



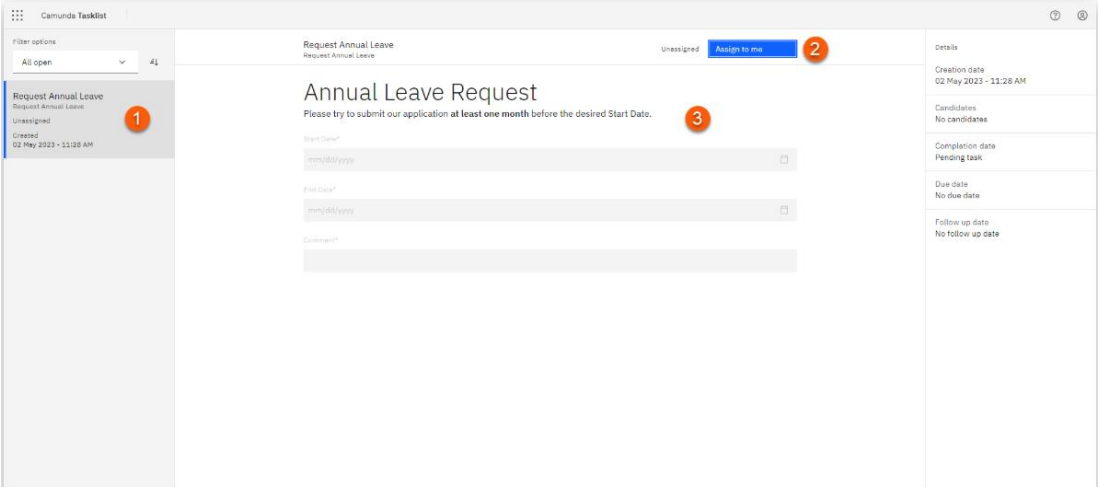
4. After a few seconds, **Tasklist** Dashboard will be displayed.
5. You will also be able to see that a *Request Annual Leave* **User Task** is visible in the left-hand panel of **Tasklist**.



Assign the Request Task

Assign the *Request Annual Leave* User Task to you by following the steps below:

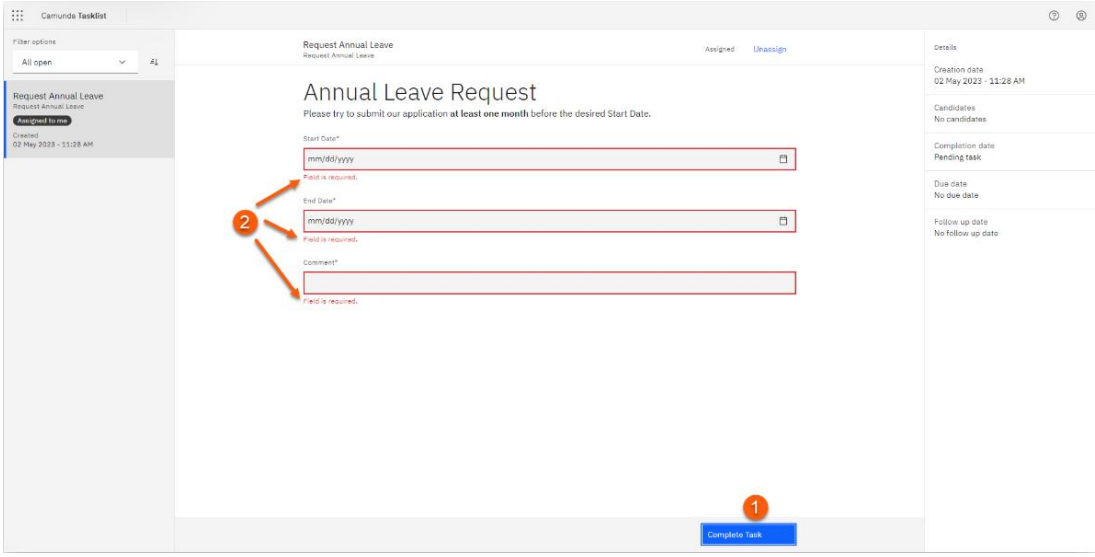
1. From the left-hand panel, click *Request Annual Leave*.
2. **Assign** the task to you.
3. You should now be presented with the **Form** that you configured in the previous steps instead of the **Add Variable** button. The **Complete Task** task button is available from the same position as before.



Complete the Request Task

Complete the *Request Annual Leave* User Task by following the steps below:

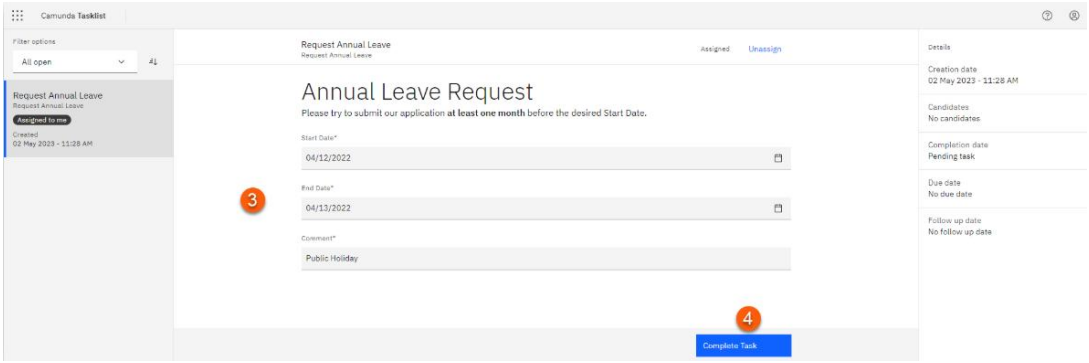
1. Click the **Complete Task** button to complete the task.
2. If you try to submit the form without adding some values you will see several validation errors due to the required fields. These validation errors will prevent you from completing the form.



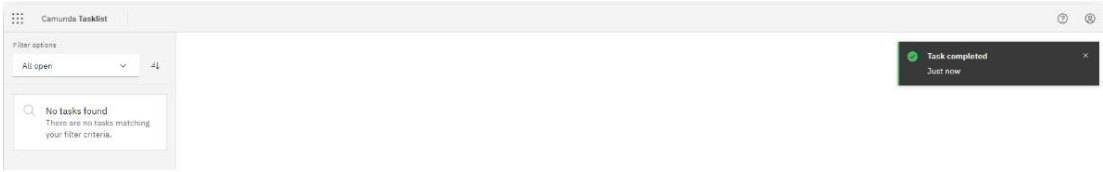
3. Now, populate the **Form** using some valid values. For example:

Name	**Value**
Start Date	04/12/2022
End Date	04/13/2022
Comment	Public Holiday

4. Click the **Complete Task** button to complete the task.



5. A **Task complete** message will be displayed at the bottom of **Tasklist** to show that this was successful.



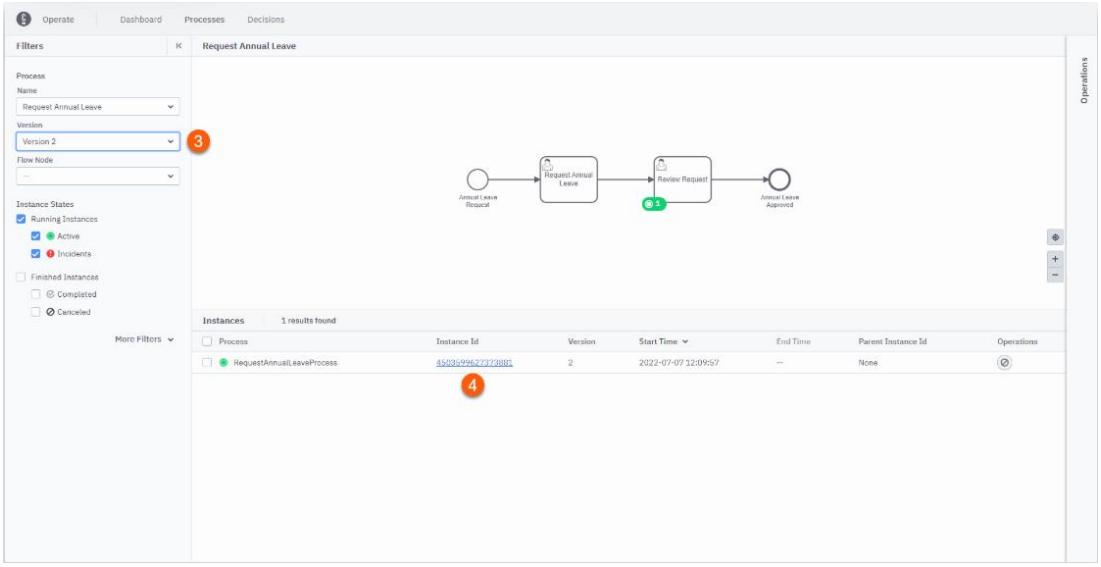
View Completed Task in Operate

We can now view the completed task in **Operate** by following the steps below:

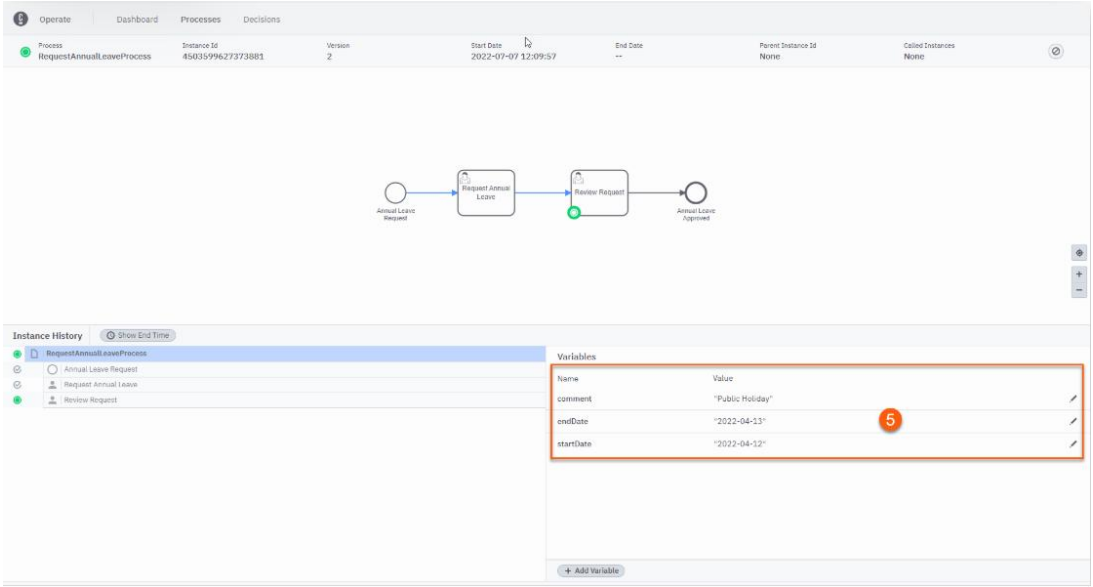
1. Launch **Operate** by following the steps in the previous lesson. Alternatively, refresh the window if you still have **Operate** open.
2. Click on *Request Annual Leave* from the **Instances by Process** panel on the left-hand side of the screen
3. **Operate** will now show a running instance of our *Request Annual Leave* process. As we now have multiple versions of the **Process Definition**, you will need to select the version number from the menu on the left-hand side of the screen.

   How are Process Instance versions managed?

   All **Process Instances** are started using the latest deployed version of the **Process Definition**. If the **Process Definition** is updated whilst a **Process Instance** is running it will continue running against the original version of the **Process Definition** that was used to start it.
4. Click on the **Instance Id** link in the **Instances** panel to view the details of the process instance.



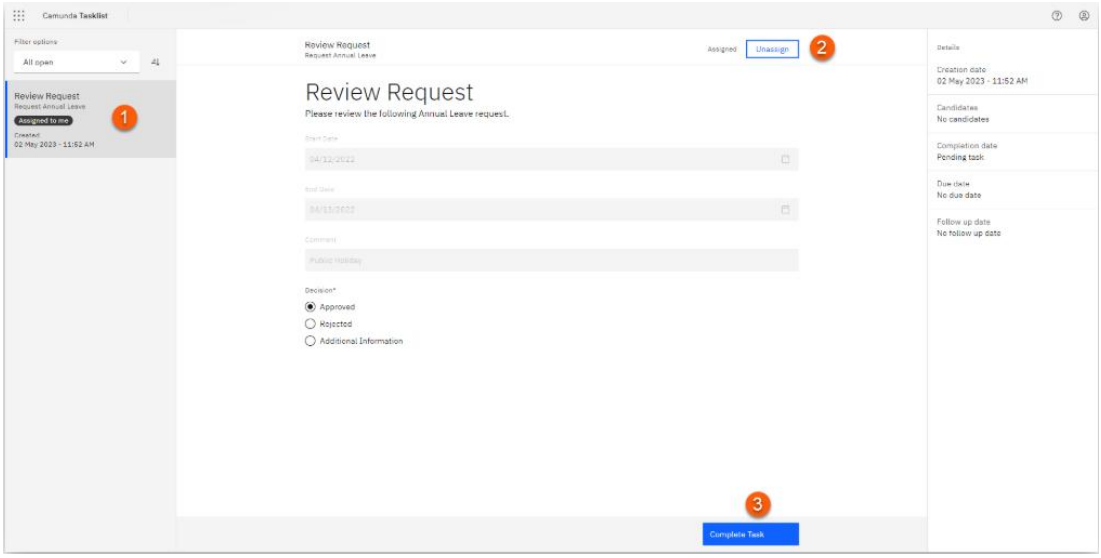
5. Note that the variables have now been populated for the process instance and that a green marker has appeared next to the *Review Request* task.



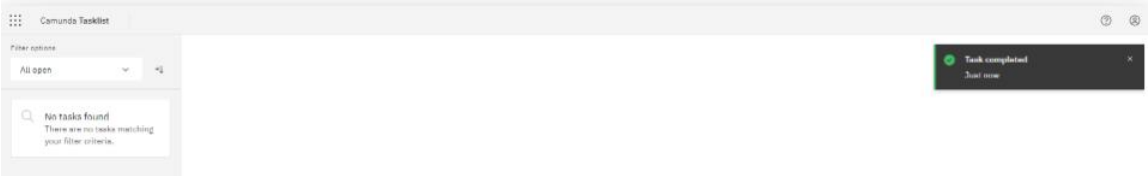
Complete the Review Task

Finally, the **Line Manager** needs to approve the request.

1. Refresh **Tasklist** to see a new *Review Request* task.
2. **Assign** the task to you. You will notice that the *Start Date*, *End Date* and *Comment* fields have variables have been populated with the previously submitted values. It is also not possible to edit these values as we set the fields to *Disabled*.
3. Select *Approved* from the *Decision* field.
4. Click the **Complete Task** button to complete the task.



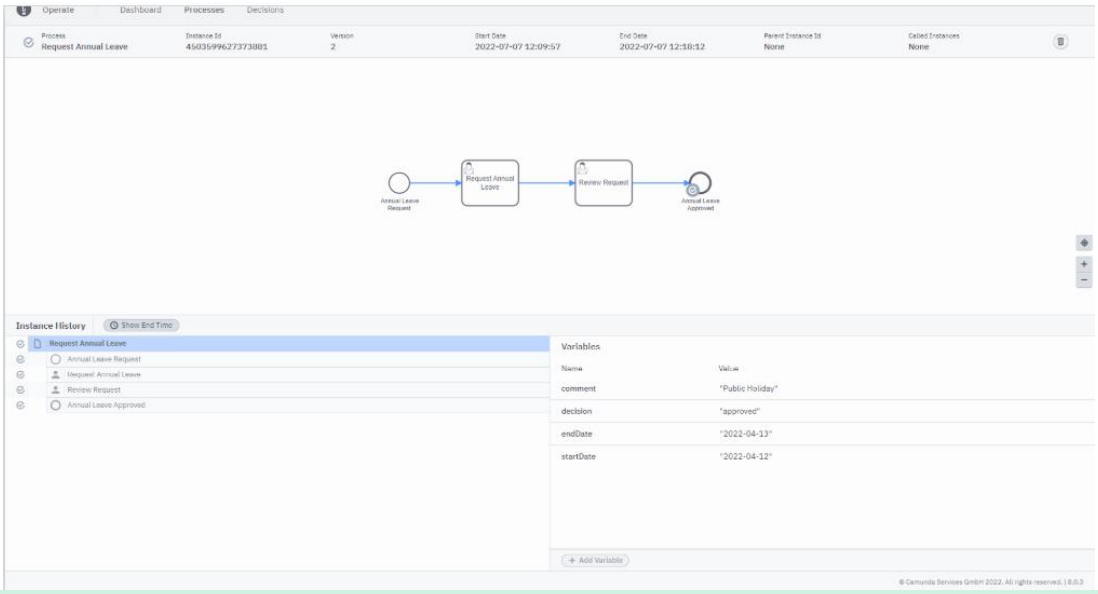
5. A **Task Complete** message will be displayed at the bottom of **Tasklist** to show that this was successful.



View Completed Review in Operate

We can now view the completed task in **Operate** by following the steps below:

1. Launch **Operate** by following the steps in the previous lesson. Alternatively, refresh the window if you still have **Operate** open.
2. Click on *Request Annual Leave* from the **Instances by Process** panel on the left-hand side of the screen
3. **Operate** will now show a completed instance of our *Request Annual Leave* process.
4. Click on the **Instance Id** link in the **Instances** panel to view the details of the process instance.
5. Note that the *Decision* variable has now been populated for the process instance and that a grey marker has appeared next to the *End Event*.



Completed a Process Instance using Forms

Congratulations! You have now completed the an instance of the *Request Annual Leave* process using forms.

Create Pool & Lanes

Overview

In the version of the process that we have developed, the decision that the **Line Manager** makes has no effect as the process always ends with *Annual Leave Approved*.

We will now configure the process logic so that the decision made by the **Line Manager** is processed by the application. This will include the following changes:

- Introduction of a **Pool** and **Lanes** to clearly show the steps performed by the **Employee** and **Line Manager**
- When the request is *Approved*, the annual leave is logged in the system and the **Employee** is notified
- When the request is *Rejected*, the **Employee** is notified
- When the request requires *Additional Information* it is sent back to the **Employee**

We will now implement the first of these changes which is the introduction of a **Pool** and **Lanes**.

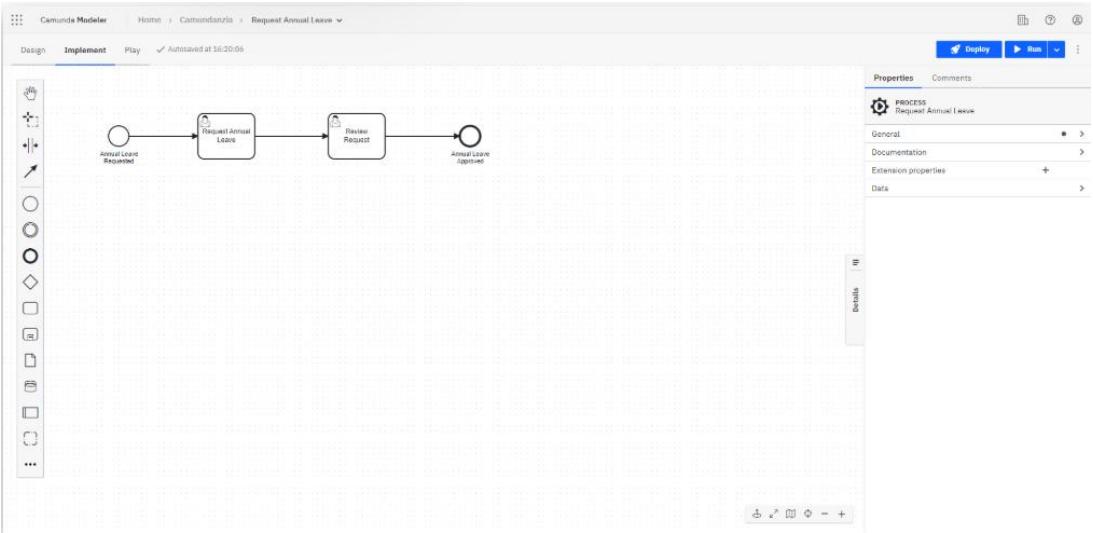
What are Pools and Lanes?

Pools and **Lanes** can be used in BPMN to clearly show who is responsible for a Task.

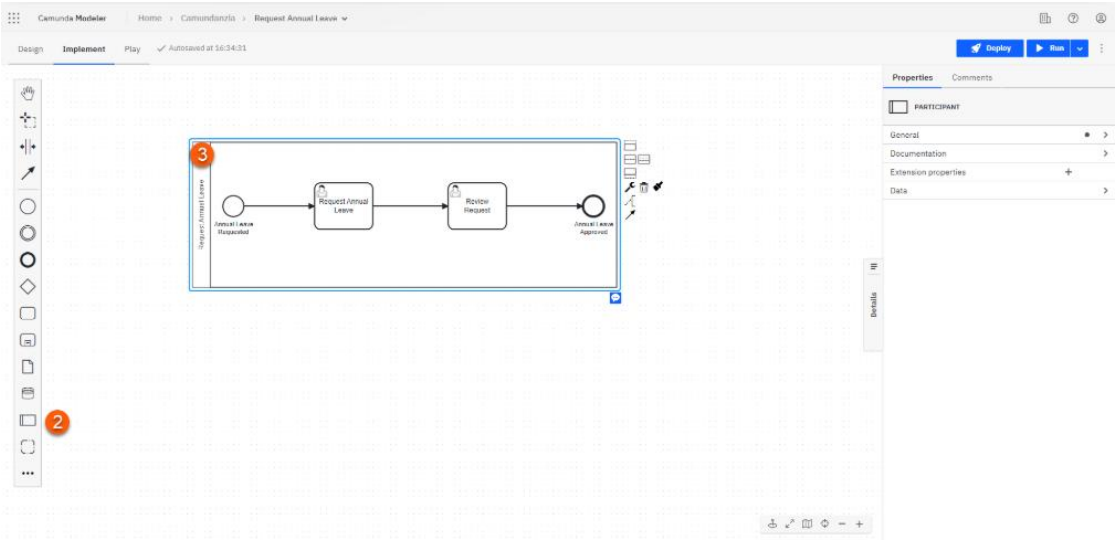
Add Pool

Create a **Pool** by following the steps below:

1. From the *Camundanzia* project in the **Modeler**, open the existing *Request Annual Leave* process



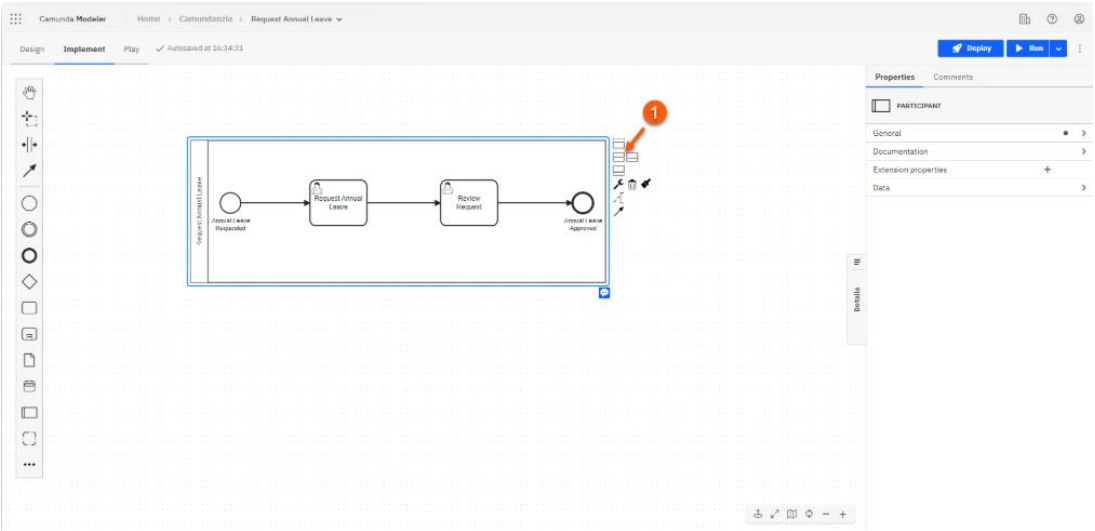
2. Click on the **Create Pool / Participant** from the **BPMN Palette** to add a new **Pool** to the diagram.
3. Name the pool *Request Annual Leave* by clicking in the vertical label.
4. Your diagram should now look similar to the below:



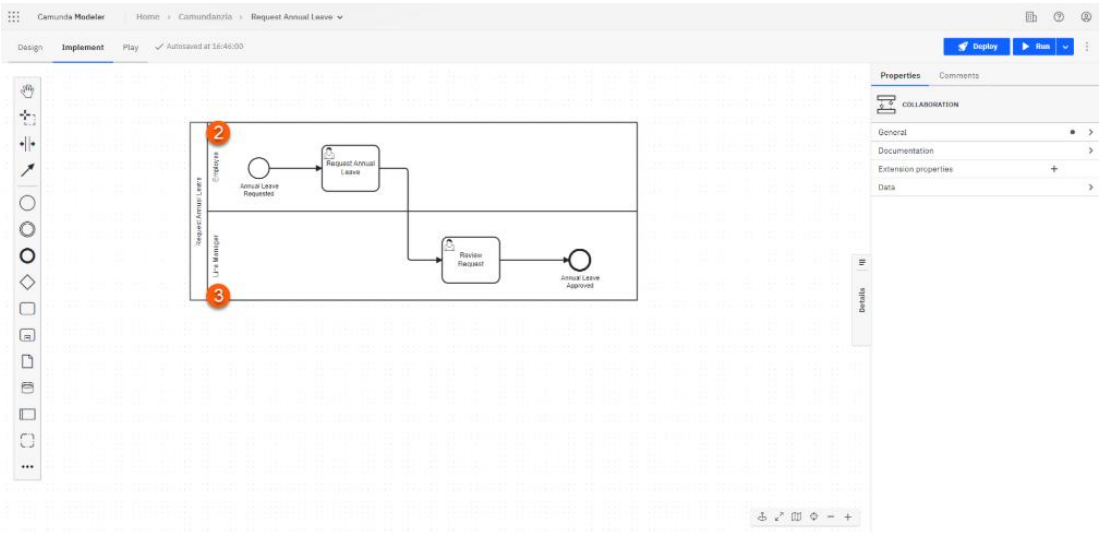
Add Lanes

Configure the **Lanes** for the process by following the steps below:

1. Click on the newly created *Request Annual Leave* **Pool** and then click on the **Divide into Two Lanes** icon from the toolbox.

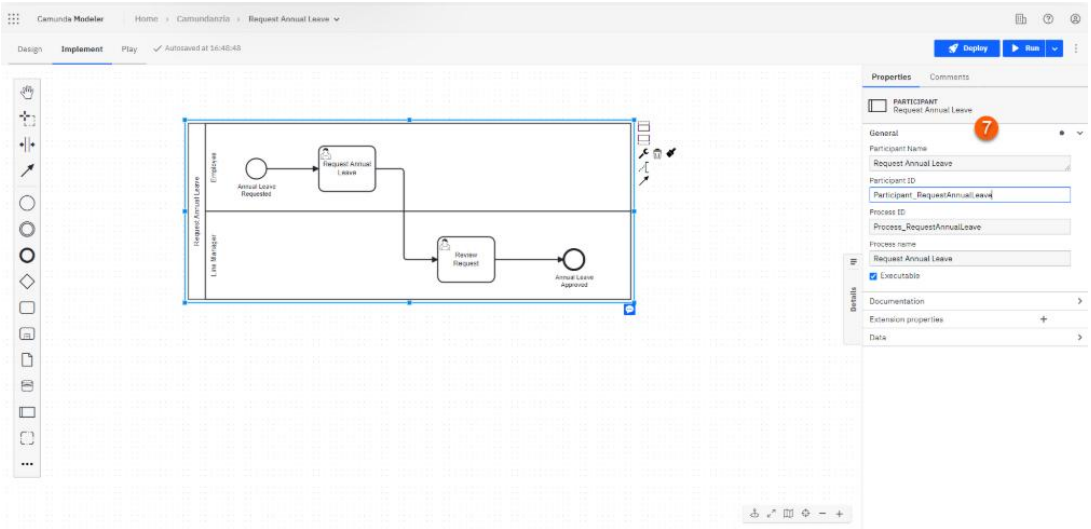


2. Name the first lane *Employee* by clicking on the left-hand side of the lane.
3. Name the second lane *Line Manager* by clicking on the left-hand side of the lane.
4. Re-arrange the diagram so that the activities that are performed by the **Employee** are shown in the *Employee* lane.
5. Re-arrange the diagram so that the activities that are performed by the **Line Manager** are shown in the *Line Manager* lane.
6. Your diagram should now look similar to the below:



7. Finally, we need to update the properties of our **Pool** so that it can be executed. Click anywhere on the **Pool** and then use the **Properties Panel** to set the following:

Name	Value
Participant Name	Request Annual Leave
Participant ID	Participant_RequestAnnualLeave
Process ID	Process_RequestAnnualLeave
Process Name	Request Annual Leave
Executable	True



User Task Assignments

To make sure the task is assigned to the correct person or group, it is possible to define an assignee, candidate users or candidate groups. Read more about User Task Assignments.

Improved Process Readability

Congratulations! You have now improved the readability of your **Process Definition** by introducing **Pools** and **Lanes**.

Add Gateway

We will now implement the second stage of our changes which is the introduction of an **Exclusive Gateway** and an additional **User Task**.

What is an Exclusive Gateway?

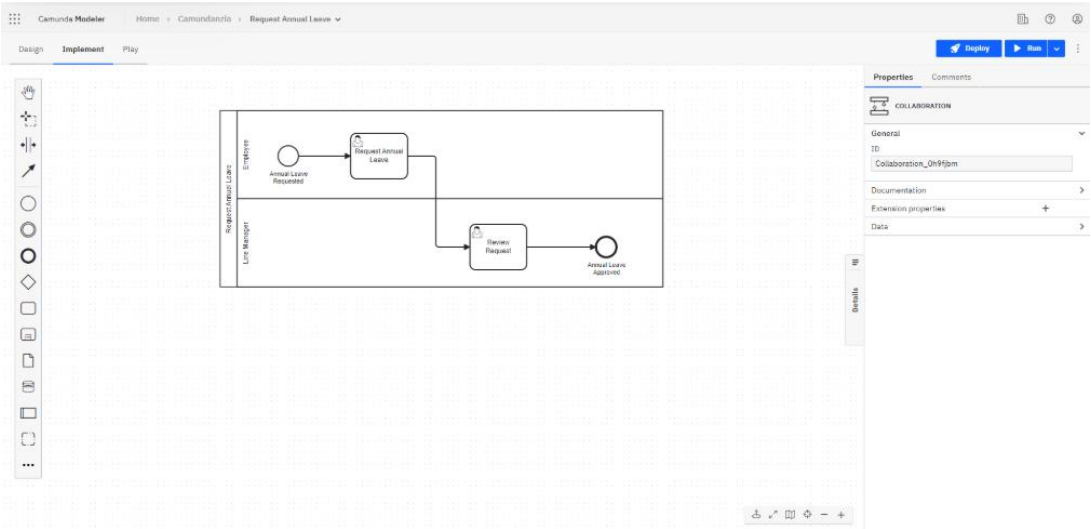
An **Exclusive Gateway** uses data to determine the next **Sequence Flow** to take in the process.

Add Additional BPMN Constructs

Add Exclusive Gateway

Add an **Exclusive Gateway** by following the steps below:

1. From the *Camundanzia* project in the **Modeler**, open the existing *Request Annual Leave* process



2. Click on the **Create Gateway** icon from the **BPMN Palette** to add a new **Exclusive Gateway** to the diagram. This should be placed after *Review Request*.

   Exclusive Gateway

   The **Exclusive Gateway** is the default gateway that is added to a diagram when using the **Create Gateway** icon.

3. Set the **Name** and **ID** for the **Gateway** as follows:

Name	Value
Name	Decision?
ID	Gateway_Decision

Add User Task

Add a **User Task** by following the steps below:

1. Click on the **Create Task** icon from the **BPMN Palette** to add a new **Task** to the diagram. This should be placed above *Review Request*.
2. Click on the **Change Type (Wrench)** icon in the toolbox and then select **User Task** from the context menu that appears
3. Set the **Name** and **ID** for the **User Task** as follows:

Name	Value
Name	Provide Additional Information
ID	Task_ProvideAdditionalInformation

Rename End Event

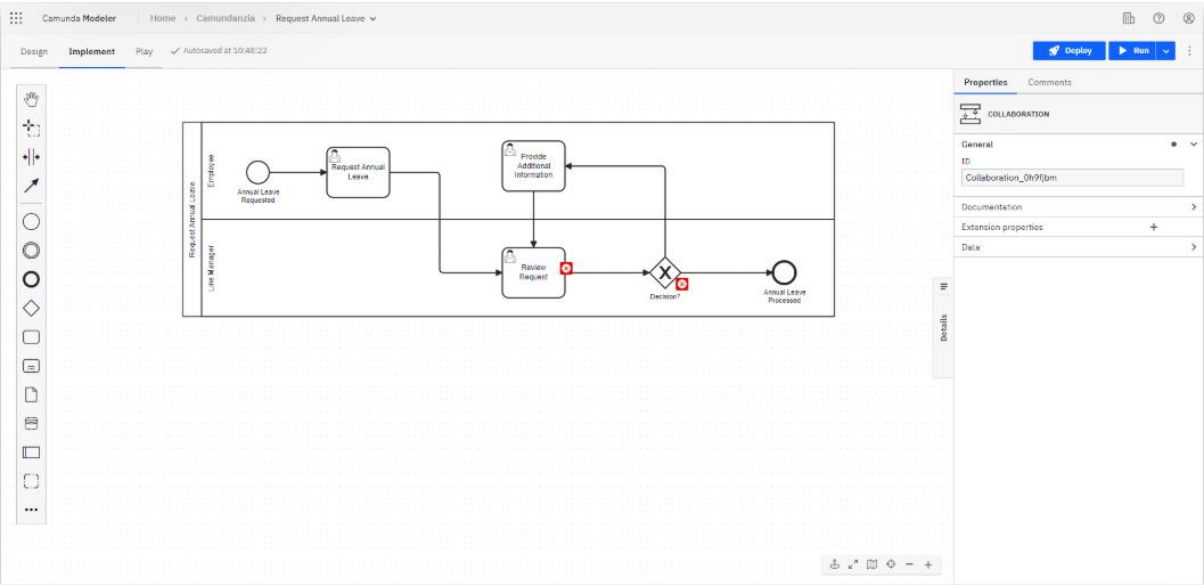
Our existing **End Event** is named to reflect the approval of the request. Let's update that to something more appropriate.

1. Click on the existing **End Event** in the diagram and set the **Name** and **ID** as follows:

Name	Value
Name	Annual Leave Processed
ID	EndEvent_AnnualLeaveProcessed

Connect Sequence Flows

Finally, re-arrange the diagram using the **Hand Tool** and the **Add / Remove Space Tool** before connecting all the **Sequence Flows** so that your diagram looks like the one below.



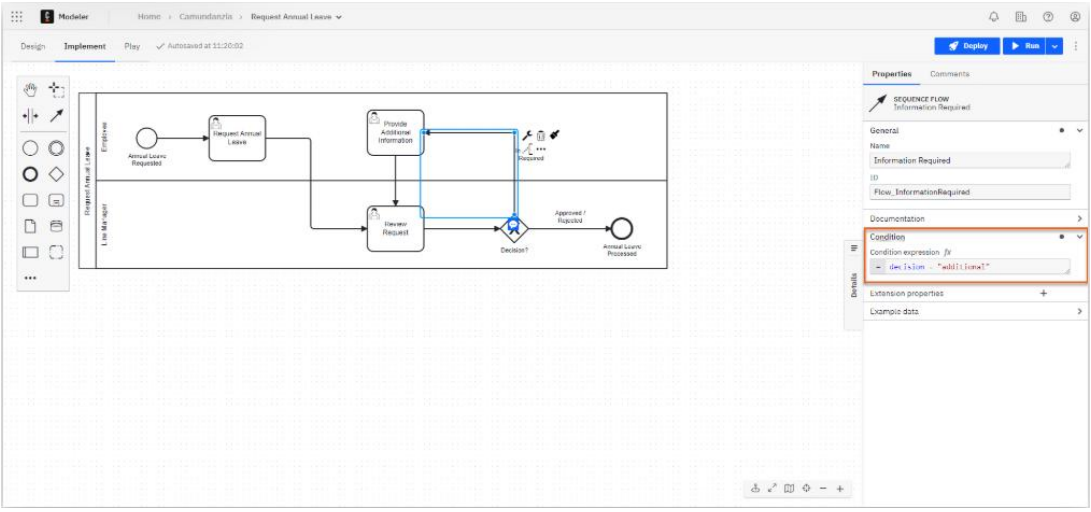
Configure Additional BPMN Constructs

Configure Exclusive Gateway

In order to make the process executable again, we need to implement the **Exclusive Gateway**. For this, we need to set the condition on each of the outgoing sequence flows from the gateway.

1. Click on the **Sequence Flow** leading from *Decision?* to *Provide Additional Information* and use the **Properties Panel** on the right-hand side of the screen to set the following:

Name	Value
Name	Information Required
ID	Flow_InformationRequired
Condition	decision = "additional"



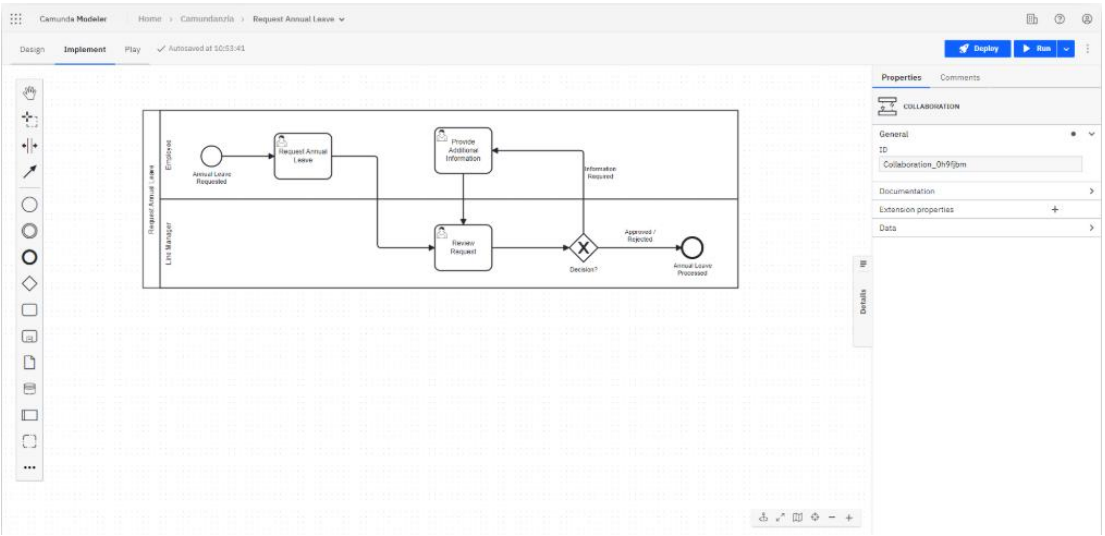
2. Click on the **Sequence Flow** leading from *Decision?* to *Annual Leave Processed* and use the **Properties Panel** on the right-hand side of the screen to set the following:

Name	**Value**
Name	Approved / Rejected
ID	Flow_ApprovedRejected
Condition	decision = "approved" or decision = "rejected"

Decision Values

The values that are set for the condition must exactly match the values specified for the radio element in the user task *Review Request*.

3. Your diagram should now look like the one below.



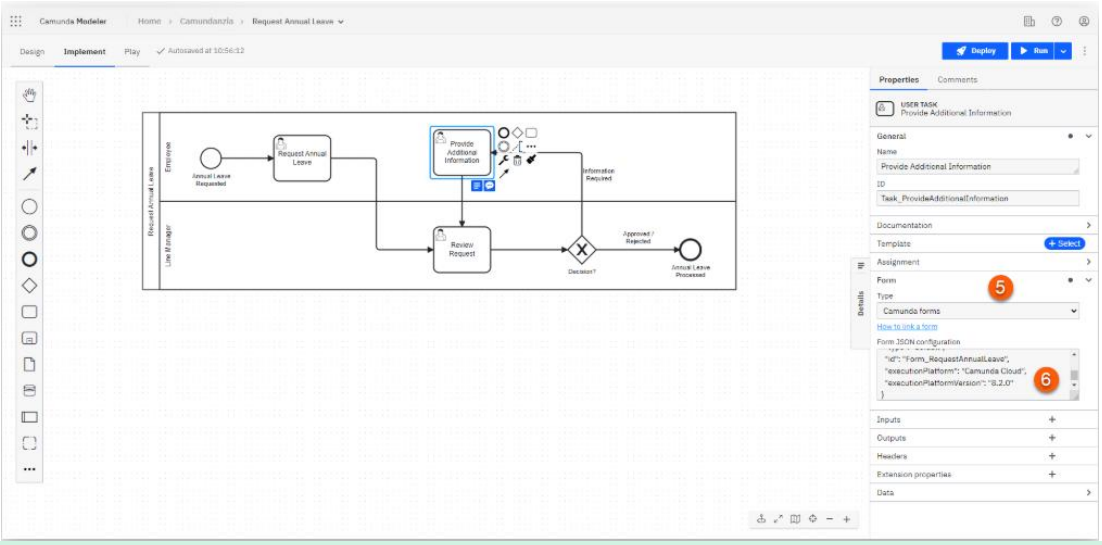
Configure User Task

We now need to configure the *Provide Additional Information* User Task so that the **Employee** can provide a comment. We can re-use our existing *Request Annual Leave* **Form** here as we are displaying and editing the same information.

Select Form

You can select the form through the blue icon or by following the steps below.

1. From the *Camundanzia* project in the **Modeler**, click the *Request Annual Leave* **Form** to launch the **Form Editor**.
2. Switch to the **Code Editor** using the **Open Code Editor** icon in the **Modeler** toolbar
3. Copy the **Form Definition** into the clipboard by using the **Copy JSON** icon in the **Modeler** toolbar
4. From the *Camundanzia* project in the **Modeler**, click the *Request Annual Leave* **BPMN Diagram** to launch the **Diagram Editor**.
5. Click the *Provide Additional Information* **User Task**, then on the right side of the screen, expand the **Form** section from the **Properties Panel**.
6. Select In the **Type** field select *Camunda forms* and then paste in the copied JSON configuration into the **Form JSON Configuration**.



Added Process Routing

Congratulations! You have now updated your process so that it routes correctly based upon the decision made by the **Line Manager**.

Add Service Implementation

We will now implement the final stage of our changes. To keep things simple, we will use **Manual Tasks** as a substitute for the *Log Request in System* and *Notify Employee* requirements. In a real application, these steps would be implemented using **Service Tasks** which would require some development work.

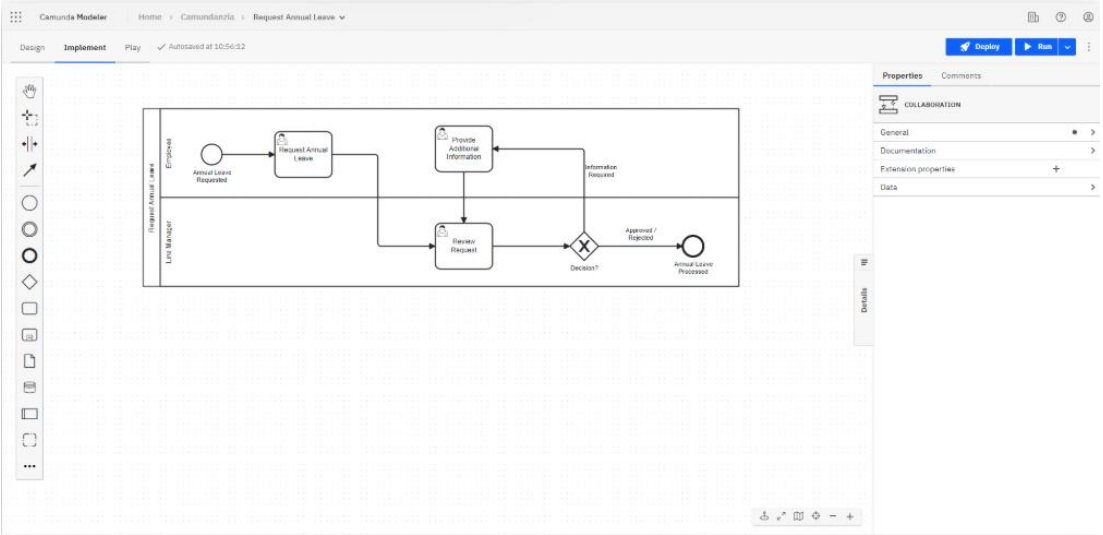
What is a Manual Task?

An **Manual Task** defines a task that is external to the **Zeebe Engine**. It is used to model work that is done by somebody who the engine does not need to know of and that has no known system or UI interface. For the engine, a **Manual Task** is handled as a pass-through activity, automatically continuing the process when the process execution arrives at it.

Add Manual Tasks

Add the **Manual Tasks** by following the steps below:

1. From the *Camundanzia* project in the **Modeler**, open the existing *Request Annual Leave* process



2. Click on the **Create Task** from the **BPMN Palette** to add a new **Task** to the diagram. This should be placed before *Annual Leave Processed*.
3. Click on the **Change Type (Wrench)** icon in the toolbox and then select **Manual Task** from the context menu that appears.
4. Set the **Name** and **ID** for the **Manual Task** as follows:

Name	Value
Name	Notify Employee
ID	Task_NotifyEmployee

5. Repeat steps 2 - 4 to create an additional **Manual Task** using the configuration below. This should be placed before *Notify Employee*.

Name	Value
Name	Log Request
ID	Task_LogRequest

Add Exclusive Gateway (Merging)

Add another **Exclusive Gateway** by following the steps below:

1. Click on the **Create Gateway** from the **BPMN Palette** to add a new **Exclusive Gateway** to the diagram. This should be placed after *Log Request* and before *Notify Employee*.
2. Set the **Name** and **ID** for the **Gateway** as follows:

Name	Value
Name	

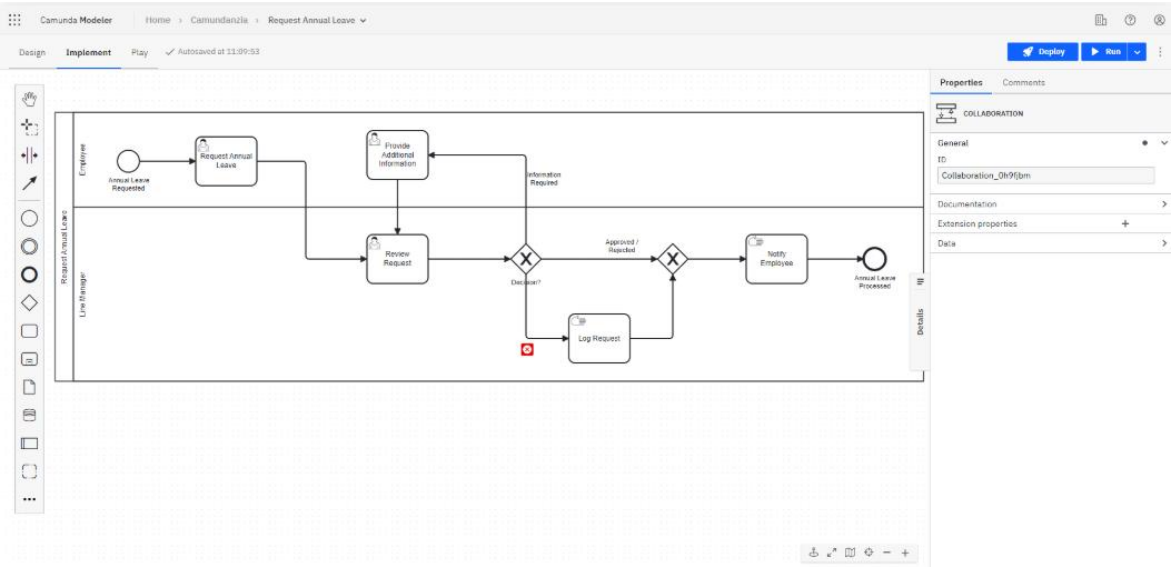
Name	Value
ID	Gateway_MergeDecision

Joining Sequence Flows with Exclusive Gateways (XOR)

Model joining the process flow by explicitly showing the **Exclusive Gateway** symbol so the reader does not need to fully understand BPMN in order to understand how two incoming or outgoing sequence flows in a task behave.

Connect Sequence Flows

Re-arrange the diagram using the **Hand Tool** and the **Add / Remove Space Tool** before connecting all the **Sequence Flows** so that your diagram looks like the one below.



Configure Exclusive Gateway

In order to make the process executable again, we need to re-configure the **Exclusive Gateway**. For this, we need to set the condition on each of the outgoing sequence flows from the gateway.

1. Click on the **Sequence Flow** leading from *Decision?* to the new **Exclusive Gateway** and use the **Properties Panel** on the right-hand side of the screen to set the following:

Name	Value
Name	Rejected
ID	Flow_Rejected
Condition	decision = "rejected"

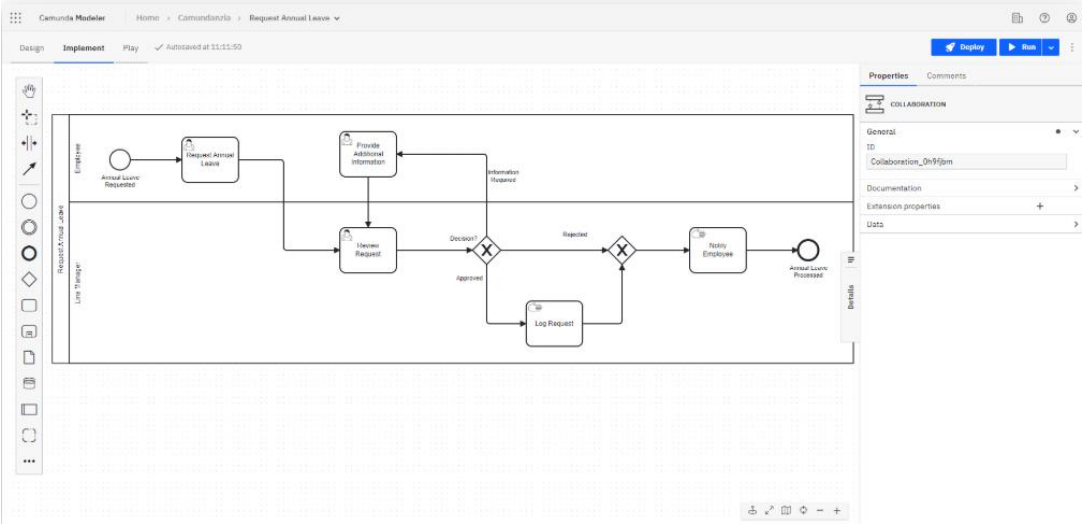
2. Click on the **Sequence Flow** leading from *Decision?* to *Log Request* and use the **Properties Panel** on the right-hand side of the screen to set the following:

Name	Value
Name	Approved
ID	Flow_Approved
Condition	decision = "approved"

Decision Values

The values that are set for the condition must exactly match the values specified for the radio element in the user task *Review Request*.

3. Your diagram should now look like the one below.



Completed Process

Congratulations! You have now updated your process so that it meets all of the requirements defined in our **Scenario**.

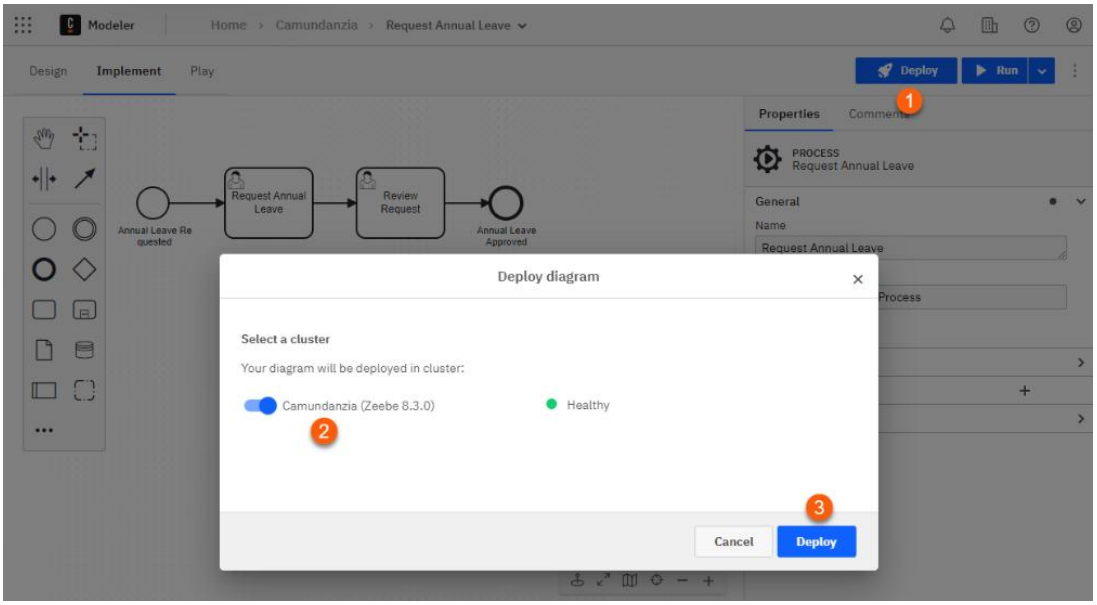
Deploy Process

We will now need to deploy our final **Process Definition** to the previously created **Cluster** so that we can start a new **Process Instance** against the updated version of the **Process Definition**.

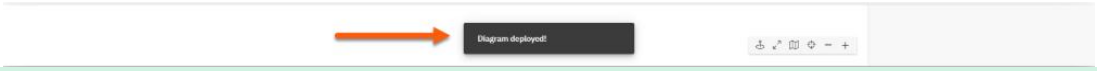
Deploy Process Definition

Deploy the **Process Definition** by following the steps below:

1. Click on the **Deploy** button in the top right corner of the **Modeler**
2. Ensure that the *Camundanzia* **Cluster** is selected from the dialog box that appears.
3. Click the **Deploy** button to deploy the **Process Definition** to the **Cluster**



4. A **Diagram Deployed** message will be displayed at the bottom of the **Modeler** to show that this was successful.



Deployed an updated Process Definition

Congratulations! You have now deployed the final version of the *Request Annual Leave* process.

Cleanup Environment

Overview

We have now met all of the business requirements that have been defined in our **Scenario**. We are now in a position where we can perform an end-to-end test of our process to ensure that everything is working as expected.

However, as we have been testing our process definition as we have been developing it, it is likely that we have several incomplete process instances still running. Having incomplete (and possibly incorrect) process instances running in our environment may make it more difficult to confirm that everything is working as expected in an end-to-end test.

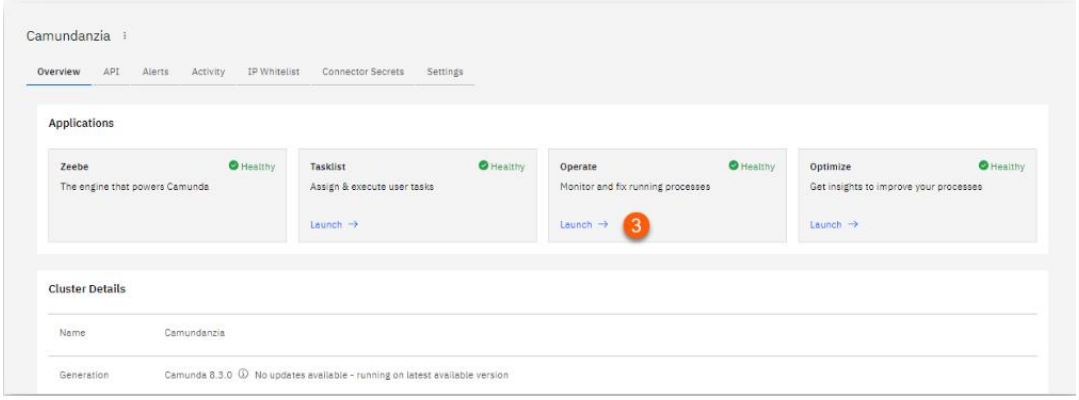
We will now **Cancel** and **Delete** any remaining instances so that we have a completely clean environment in which we can conduct our acceptance tests.

Cancel & Delete Process Instances

Launch Operate

Launch **Operate** by following the steps below:

1. From the **Console**, select **Clusters**.
2. Select *Camundanzia* from the **Clusters** page.
3. Click the **Launch** link next to **Operate** in the **Cluster Details** page



4. After a few seconds, the **Operate** Dashboard will be displayed.

Cancel Process Instances

We can now **Cancel** the process instances in **Operate** by following the steps below:

1. Click on *Request Annual Leave* from the **Instances by Process** panel on the left-hand side of the screen
2. On the bottom left of the screen is a series of filters that will allow you to select *Running* or *Finished* process instances.
3. Ensure that the *Running* filters are selected and the *Finished* filters are not selected.
4. Then, for each **Process Instance** that is shown in the **Instances** panel, click the **Cancel Instance** button to the right-hand side of the process.

No Running Processes

This section can be skipped if there are no running processes shown in the **Instances** panel.

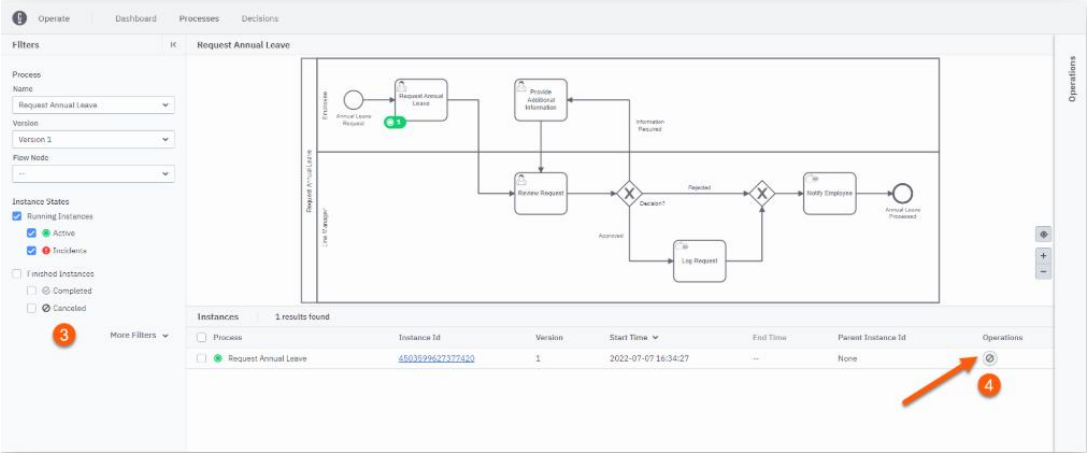
Multiple Process Definitions

You should see two Process Definitions in Operate. The first is the original process without a pool, the second is the new process which uses pools and lanes. Further revisions to the process will result in new versions of the existing process rather than a new process definition.

5. The **Process Instance** will then be cancelled and will disappear from the **Instances** panel.

What happened to the Process Instance?

When you **Cancel** a process in Operate it changes to a stopped state. As the process is no longer running it is not picked up by *Running* filter.



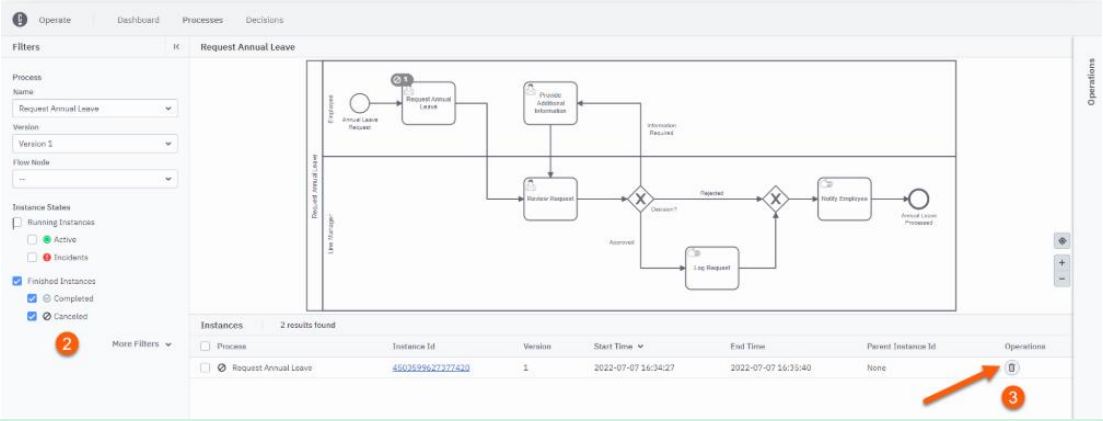
Delete Process Instances

We can now **Delete** the finished process instances in **Operate** by following the steps below:

1. On the bottom left of the screen is a series of filters that will allow you to select *Running* or *Finished* process instances.
2. Ensure that the *Finished* filters are selected and the *Running* filters are not selected.
3. Then, for each **Process Instance** that is shown in the **Instances** panel, click the **Delete Instance** button to the right-hand side of the process.
4. The **Process Instance** will then be deleted and will disappear from the **Instances** panel.

What happened to the Process Instance?

When you **Delete** a process in Operate it is completely removed from the **Zeebe Engine**.



Cleaned the Environment

Congratulations! You have now removed all instances of *Request Annual Leave* from the environment.

Test Process

We are now ready to test the *Request Annual Leave* process.

Review Acceptance Criteria

Let's start by reviewing the acceptance criteria that was defined earlier; we need to ensure that all of the requirements below can be met by our process.

1. **Employees** are able to submit an annual leave request
2. **Line Managers** can review annual leave requests

3. **Line Managers** can approve, reject or ask for more details on a request
4. **Employees** can provide additional information on a request and then resubmit
5. **Line Managers** can review additional information on a request before making a decision

Test Application

Test Scenarios

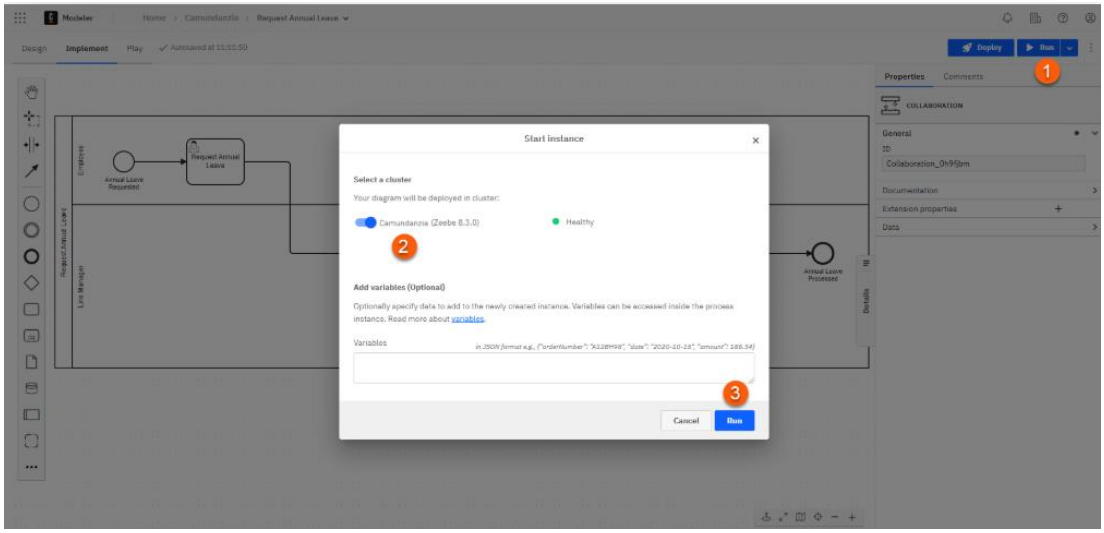
Three scenarios will be required to confirm that the acceptance criteria above has been met.

1. Annual Leave Request is *Approved*
2. Annual Leave Request is *Rejected*
3. Annual Leave Request requires *Additional Information* and is subsequently *Approved*

Launch Process Instances

Start **three instances** of the *Request Annual Leave* process by following the steps below:

1. Click on the **Run** button in the top right corner of the **Modeler**
2. Ensure that the *Camundanzia* **Cluster** is selected from the dialog box that appears.
3. Click the **Run** button to start a new instance of *Request Annual Leave* using the *Camundanzia* **Cluster**

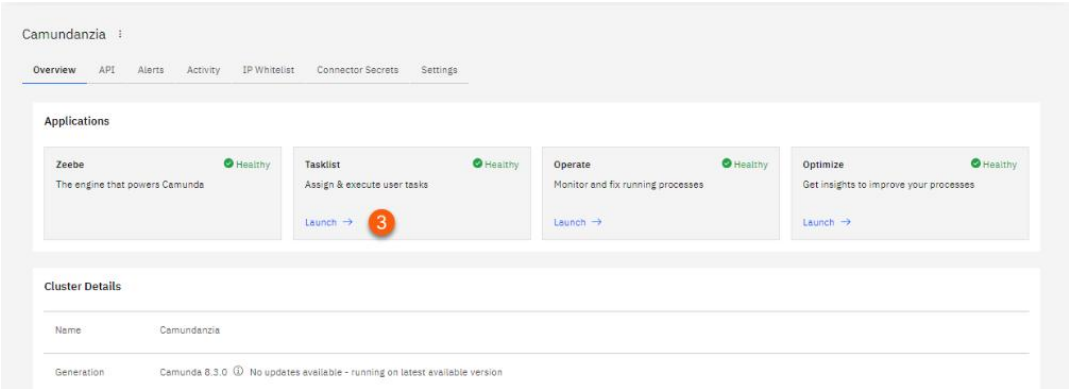


4. An **Instance Started!** message will be displayed at the bottom of the **Modeler** to show that this was successful.

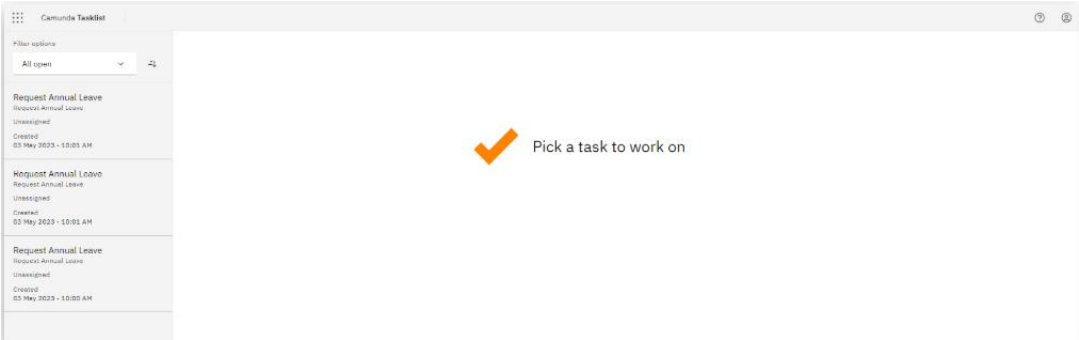
Create the Requests

Create the requests in **Tasklist** by following the steps below:

1. From the **Console**, select **Clusters**.
2. Select *Camundanzia* from the **Clusters** page.
3. Click the **Launch** link next to **Tasklist** in the **Cluster Details** page



4. After a few seconds, the **Tasklist** Dashboard will be displayed.
5. You will also be able to see that three *Request Annual Leave* tasks are visible in the left-hand panel of **Tasklist**.



6. **Assign to you** and **Complete** each of the three requests using the following values:

Test Scenario	Name	Value
1 - Approved	Start Date	01/01/2023
1 - Approved	End Date	01/02/2023
1 - Approved	Comment	Test Scenario - Approved
2 - Rejected	Start Date	01/01/1999
2 - Rejected	End Date	01/02/1999
2 - Rejected	Comment	Test Scenario - Rejected
3 - Additional	Start Date	01/01/2023
3 - Additional	End Date	04/01/2023
3 - Additional	Comment	Test Scenario - Additional

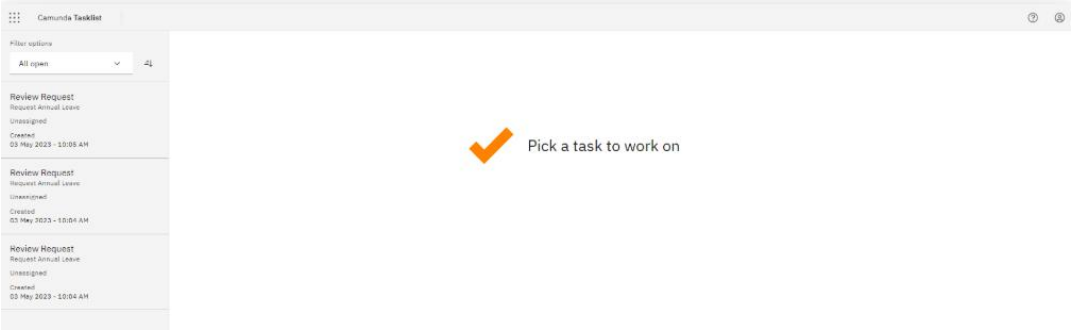
Acceptance Criteria - Employees are able to submit an annual leave request

Congratulations! You have now demonstrated that the first item of the **Acceptance Criteria** has been met. Employees are able to submit an annual leave request.

Review the Requests

Review the requests in **Tasklist** by following the steps below:

1. Refresh the **Tasklist** dashboard.
2. You should now see three *Review Request* tasks in the left-hand panel of **Tasklist**.



3. **Assign to you** and **Complete** each of the three requests using the following values:

Test Scenario	Decision
1 - Approved	Approved
2 - Rejected	Rejected
3 - Additional	Additional Information

Acceptance Criteria - Line Managers can review annual leave requests

Congratulations! You have now demonstrated that the second item of the **Acceptance Criteria** has been met. Line Managers can review annual leave requests.

Acceptance Criteria - Line Managers can approve, reject or ask for more details on a request

Congratulations! You have now demonstrated that the third item of the **Acceptance Criteria** has been met. Line Managers can approve, reject or ask for more details on a request.

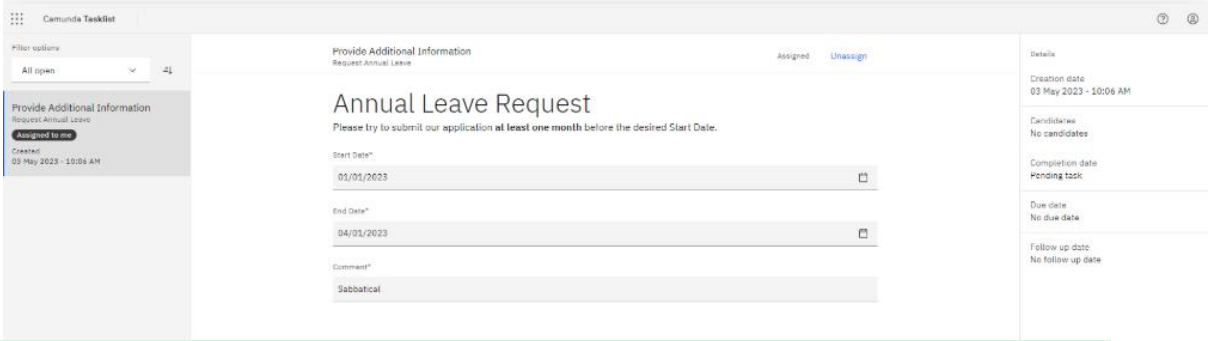
Provide Additional Information

As the **Employee**, provide some additional information for the request which was returned by the **Line Manager**.

1. Refresh the **Tasklist** dashboard.
2. You should now see a single *Provide Additional Information* task in the left-hand panel of **Tasklist**.
3. **Assign to you** and **Complete** the task using the following values:

Test Scenario	Comment

Test Scenario	Comment
3 - Additional	Sabbatical



Acceptance Criteria - Employees can provide additional information on a request and then resubmit

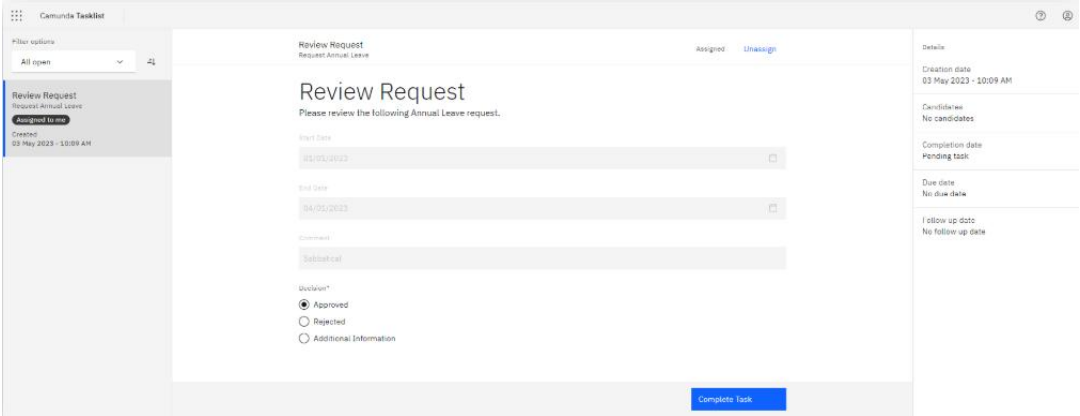
Congratulations! You have now demonstrated that the fourth item of the **Acceptance Criteria** has been met. Employees can provide additional information on a request and then resubmit.

Review Additional Information

As the **Line Manager**, review the additional information for the request which was returned by the **Employee**.

1. Refresh the **Tasklist** dashboard.
2. You should now see a single *Review Request* task in the left-hand panel of **Tasklist**.
3. **Assign to you** and **Complete** the task using the following values:

Test Scenario	Decision
3 - Additional	Approved



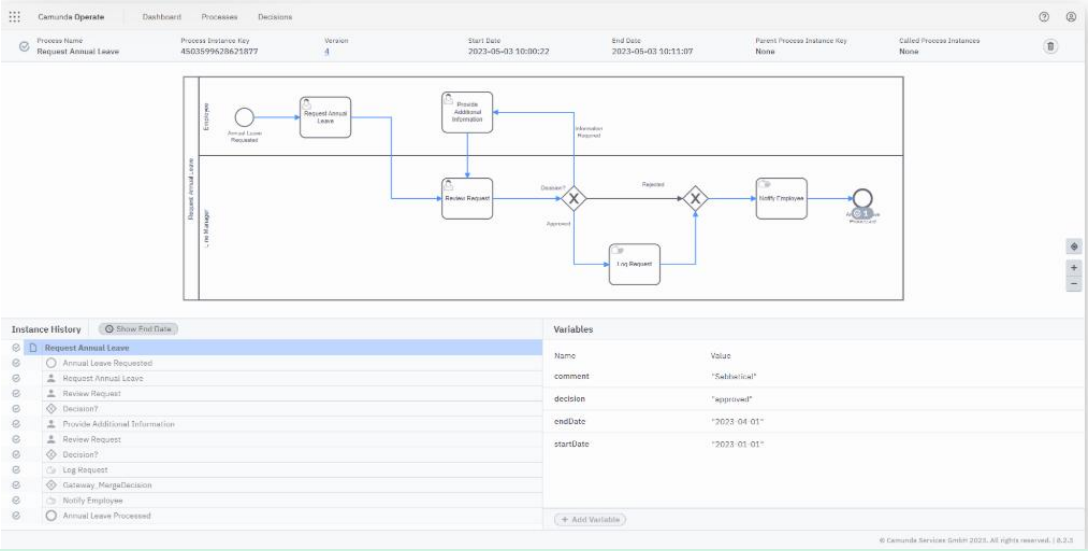
Acceptance Criteria - Line Managers can review additional information on a request before making a decision

Congratulations! You have now demonstrated that the final item of the **Acceptance Criteria** has been met. Line Managers can review additional information on a request before making a decision.

View Completed Request Task in Operate

Finally, we can now view each of the completed process instances in **Operate** by following the steps below:

1. Launch **Operate** by following the steps in the previous lesson. Alternatively, refresh the window if you still have **Operate** open.
2. Click on *Request Annual Leave* from the **Instances by Process** panel on the left-hand side of the screen
3. Ensure that the *Finished* filters are selected and the *Running* filters are not selected.
4. Then, for each **Process Instance** that is shown in the **Instances** panel, click on the **Instance Id** link in the **Instances** panel to view the details of the process instance.
5. Note that the **Instance History** and **Variables** have been populated for the process instance and that the sequence of activities follows our test scenario.



Success

Congratulations! You have now completed the testing of your process and have confirmed that it meets the acceptance criteria defined at the start of this course.

Example Solution

If you encountered any problems during this course or were unable to complete the process configuration, you can compare your solution with our example solution below.

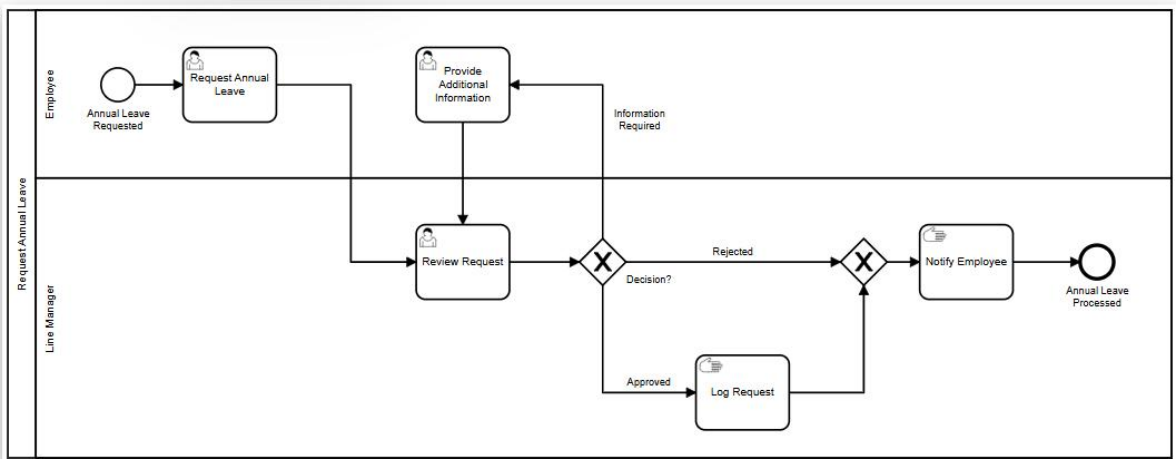
GitHub Repository

Our version of the **Process Definition** including the **Form** definitions can be found in the GitHub Repository below:

- Solution - Request Annual Leave

Process Diagram

Our version of the **Process Diagram** can be found below:


Request Annual Leave
Employee
Line Manager
Annual Leave Requested
Request Annual Leave
Provide Additional Information
Information Required
Review Request
Decision?
Rejected
Approved
Log Request
Notify Employee
Annual Leave Processed